# NITA Classes — DCA / NIELIT CCC 90-Hour Study Pack

**90 Hours** (30 Theory + 60 Practical) · **Exam:** 100 MCQ · 90 min · 50% pass

# Module 1: Introduction to Computer (10 Hours)

**Course:** CCC 90-Hour DCA | **Module duration:** 10 hours

## Learning Objectives

After this module you will be able to:

- Explain what a computer is, its applications, and types
- Describe the evolution of computers through five generations
- Identify hardware and software components (CPU, memory, I/O, storage)
- Differentiate data, information, and files
- Understand basic number systems (binary, decimal)
- Recognize modern IT gadgets (smartphone, tablet, IoT) and mobile apps
- Follow safe startup and shutdown procedures

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## 1. Introduction to Computers

**Computer** = electronic device that accepts **input**, processes it using **programs**, stores **data**, and gives **output**.

### Applications (व्यवहारिक उपयोग)

- Education, banking, hospitals, offices, e-governance
- Internet, email, online forms, DigiLocker, UPI payments
- CCC and government job exams test basic computer awareness

### Types of Computers

| Type | Size | Use |
|---|---|---|
| Microcomputer (PC, laptop) | Small | Home, office, institute |
| Minicomputer | Medium | Departments, servers |
| Mainframe | Large | Banks, railways |
| Supercomputer | Very large | Weather, research, AI |

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## 2. Evolution of Computers — Five Generations

| Generation | Period | Technology | Key features | Examples |
|---|---|---|---|---|
| **First (1G)** | 1940s–1950s | Vacuum tubes | Very large, slow, high power | ENIAC, UNIVAC |
| **Second (2G)** | 1950s–1960s | Transistors | Smaller, faster, more reliable | IBM 1401 |
| **Third (3G)** | 1960s–1970s | Integrated circuits (ICs) | Multiprogramming, minicomputers | IBM System/360 |

| | | | | |
|---|---|---|---|---|
| **Fourth (4G)** | 1970s–present | Microprocessors (VLSI) | Personal computers, GUI, networks | Intel PCs, laptops |
| **Fifth (5G)** | Present & future | AI, parallel processing, ULSI | Smart devices, cloud, IoT, voice assistants | Smartphones, supercomputers |

**Exam tip:** Each generation is defined by its **hardware technology**, not by internet speed or mobile networks (5G mobile ≠ fifth computer generation).

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## 3. IT Gadgets and Mobile Computing

### Smartphones

- Pocket computer with **touchscreen**, **camera**, **GPS**, **mobile OS** (Android, iOS)
- **Mobile apps** = application software installed from Play Store / App Store
- Uses: calls, messaging, banking (UPI), learning apps, government services (mAadhaar, UMANG)

### Tablets

- Larger touchscreen than phone; lighter than laptop
- Good for reading, video classes, note-taking with stylus
- Examples: iPad, Samsung Galaxy Tab, budget Android tablets

### IoT (Internet of Things) Devices

- Everyday objects connected to the internet and controlled remotely
- Examples: smart bulbs, CCTV cameras, fitness bands, smart TVs, home assistants
- Need Wi-Fi and often a **mobile app** for setup and control

### Other Common Gadgets

| Gadget | Function |
|---|---|
| Laptop | Portable PC for study and office work |
| Pen drive | Portable USB storage |
| External HDD/SSD | Backup and large file storage |
| Webcam / Headset | Online classes and video calls |
| Printer / Scanner | Hard copy output and digitizing documents |

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## 4. Components of a Computer System

```
Input → CPU (Process) → Memory → Output
              ↓
           Storage
```

### Hardware (tangibles — छू सकते हैं)

| Component | Function | Examples |
| --- | --- | --- |
| **CPU** | Brain — executes instructions | Intel Core i3/i5, AMD Ryzen |
| **RAM** | Temporary memory while working | 4 GB, 8 GB, 16 GB |
| **ROM** | Permanent boot instructions | BIOS chip |
| **Hard Disk / SSD** | Permanent storage | 256 GB, 512 GB, 1 TB |
| **Motherboard** | Connects all parts | — |
| **Input devices** | Send data to computer | Keyboard, mouse, scanner, microphone |
| **Output devices** | Show/print results | Monitor, printer, speaker |
| **VDU** | Visual Display Unit = Monitor | LCD, LED |

### Software (intangible — programs)

- **System software:** Operating System (Windows, Linux, Android)
- **Application software:** Word processor, spreadsheet, browser, mobile apps
- **Utility software:** Antivirus, disk cleanup

**Firmware:** Software stored in hardware (e.g. BIOS).

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## 5. Central Processing Unit (CPU)

CPU has three parts:

1. **ALU** (Arithmetic Logic Unit) — math and comparisons
2. **CU** (Control Unit) — directs operations
3. **Registers** — tiny fast storage inside CPU

**Speed** measured in **GHz** (gigahertz). More GHz + more cores = faster processing.

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## 6. Memory

| Memory | Full Form | Volatile? | Use |
| --- | --- | --- | --- |
| RAM | Random Access Memory | Yes (clears on shutdown) | Running programs |
| ROM | Read Only Memory | No | Boot-up |
| Cache | — | Yes | Very fast buffer for CPU |
| Virtual memory | — | Uses disk as extra RAM | When RAM is full |

**1 Byte** = 8 bits
**1 KB** = 1024 bytes | **1 MB** = 1024 KB | **1 GB** = 1024 MB | **1 TB** = 1024 GB

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## 7. Input and Output Devices

### Input

- **Keyboard:** QWERTY layout; function keys F1–F12; shortcuts Ctrl, Alt, Shift
- **Mouse:** Left click (select), right click (menu), double-click (open), scroll wheel
- **Scanner:** Converts paper → digital image
- **Webcam / Microphone:** Video calls, recording
- **Touchscreen:** Input on phones and tablets

### Output

- **Monitor:** Resolution e.g. 1920×1080 (Full HD)
- **Printer:** Inkjet (colour photos), Laser (fast text), Dot matrix (bills)
- **Speaker / Headphones:** Audio output

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## 8. Storage Devices

| Device | Type | Speed | Portability |
|---|---|---|---|
| HDD | Magnetic disk | Slower | Internal/external |
| SSD | Flash chips | Fast | Internal/external |
| Pen drive (USB) | Flash | Medium | Highly portable |
| CD/DVD | Optical | Slow | Portable |
| Memory card | Flash | Medium | Camera, phone |

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## 9. Data and Information

- **Data:** Raw facts (numbers, text, images) — e.g. `85, 90, 78`
- **Information:** Processed meaningful data — e.g. "Average marks = 84.3"

**File:** Named collection of data stored on disk.
**Folder (Directory):** Container for files and subfolders.

Common extensions: `.txt`, `.docx`, `.odt`, `.xlsx`, `.ods`, `.pptx`, `.odp`, `.pdf`, `.jpg`, `.mp4`, `.apk`

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## 10. Number Systems (Basic)

| System | Base | Digits | Use |
|---|---|---|---|
| Decimal | 10 | 0–9 | Daily life |
| Binary | 2 | 0, 1 | Computer internal |
| Hexadecimal | 16 | 0–9, A–F | Memory addresses |

**Binary example:** `1010` = 1×8 + 0×4 + 1×2 + 0×1 = **10** (decimal)

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## 11. Connecting a Computer (Lab Theory)

### Steps to set up a desktop PC

1. Place CPU, monitor, keyboard, mouse on stable surface
2. Connect monitor cable (HDMI/VGA) to CPU
3. Connect keyboard and mouse to USB ports
4. Connect power cables; switch on UPS first (if used), then CPU and monitor
5. Wait for OS logo to appear

### Safe shutdown

1. Save all work
2. Start → Power → Shut down (never unplug while running)
3. Switch off UPS after shutdown completes

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## 12. Important Terms (Exam Vocabulary)

| Term | Meaning |
|---|---|
| Hardware | Physical parts of computer |
| Software | Programs and instructions |
| Booting | Starting the computer |
| Cold boot | Start from off state |
| Warm boot | Restart without full power off |
| GUI | Graphical User Interface (icons, mouse) |
| CLI | Command Line Interface (text commands) |
| Multitasking | Running multiple programs at once |
| Mobile app | Application software on smartphone/tablet |
| IoT | Devices connected to internet for remote control |
| Virus | Harmful program that spreads |
| Backup | Copy of data for safety |

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## Lab Tasks (Hands-on)

1. **Identify parts:** Label CPU, monitor, keyboard, mouse, USB ports on your lab PC.
2. **Storage check:** Open File Manager / This PC → note C: drive total size and free space.
3. **Device list:** List 3 input and 3 output devices you use daily (include phone/tablet).
4. **File types:** Create or locate 5 files with extensions `.txt` , `.docx` , `.xlsx` , `.pptx` , `.jpg` — observe icons.
5. **Binary convert:** Convert decimal 13, 25, 31 to binary (show working).
6. **Generations chart:** Draw a table of 5 computer generations with technology used in each.
7. **Mobile apps:** On a smartphone, list 5 apps — classify each as system, utility, or application software.

8. **IoT survey:** Name 3 IoT or smart gadgets at home or college and what they connect to.

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## Chapter Summary Checklist

- [ ] I can name 5 hardware and 5 software examples
- [ ] I know all 5 generations and their key technology
- [ ] I know RAM vs ROM vs hard disk
- [ ] I can explain input vs output devices
- [ ] I know KB, MB, GB conversions
- [ ] I can describe smartphone, tablet, and IoT basics
- [ ] I can connect and safely shut down a PC

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## Quick MCQs (10 Questions)

**1.** Which generation of computers used vacuum tubes?

- a) Second
- b) Third
- c) First
- d) Fourth

**2.** The brain of the computer is:

- a) RAM
- b) CPU
- c) Hard disk
- d) Monitor

**3.** Which memory is lost when power is switched off?

- a) ROM
- b) RAM
- c) BIOS
- d) Firmware

**4.** A smartphone app downloaded from Play Store is an example of:

- a) Hardware
- b) System software
- c) Application software
- d) Firmware

**5.** IoT devices are best described as:

- a) Computers without keyboards
- b) Objects connected to the internet for monitoring/control
- c) Only supercomputers
- d) Optical storage devices

**6.** Binary number `1101` in decimal is:

- a) 11

- b) 13
- c) 15
- d) 9

**7.** Fourth generation computers are based on:

- a) Vacuum tubes
- b) Transistors
- c) Microprocessors
- d) Mechanical gears

**8.** Which is an output device?

- a) Scanner
- b) Keyboard
- c) Microphone
- d) Printer

**9.** 1 GB equals approximately:

- a) 1000 bytes
- b) 1024 MB
- c) 1024 KB
- d) 8 bits

**10.** Processed meaningful data is called:

- a) Input
- b) Output
- c) Information
- d) Hardware

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### MCQ Answers

1-c | 2-b | 3-b | 4-c | 5-b | 6-b | 7-c | 8-d | 9-b | 10-c

# Module 2: Operating System (10 Hours)

**Course:** CCC 90-Hour DCA | **Module duration:** 10 hours

## Learning Objectives

After this module you will be able to:

- Describe functions of an operating system
- Navigate Windows desktop, Start menu, taskbar, and Settings
- Create, rename, copy, move, and delete files and folders
- Use File Explorer effectively
- Install and remove (uninstall) programs safely
- Understand Linux/Ubuntu basics and open-source software
- Use built-in utilities (Notepad, Calculator, Paint, Snipping Tool)
- Apply basic security habits (lock screen, user accounts, updates)

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## 1. Introduction to Operating System

**OS** = software that manages hardware and provides interface for users and applications.

### Functions of OS

1. **Process management** — runs and schedules programs
2. **Memory management** — allocates RAM to applications
3. **File management** — folders, permissions, storage
4. **Device management** — drivers for printer, keyboard, etc.
5. **User interface** — GUI (icons, mouse) or CLI (text commands)

### Popular Operating Systems

| OS | Type | Use |
|---|---|---|
| Windows 10/11 | GUI | Most offices, institutes, CCC labs |
| Linux (Ubuntu) | GUI + CLI | Servers, developers, free PCs |
| macOS | GUI | Apple computers |
| Android / iOS | Mobile GUI | Smartphones and tablets |

This module focuses on **Windows GUI**, with **Linux/Ubuntu** and **open source** introduced for CCC syllabus coverage.

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## 2. Open Source Software and Linux/Ubuntu

### Open Source

- **Source code** is publicly available — anyone can view, modify, and share (under a license)
- Usually **free to download** and use; community or foundation maintains it

- Examples: **Linux**, **LibreOffice**, **Firefox**, **VLC Media Player**

### Closed Source (Proprietary)

- Source code is private; only the vendor can modify
- Examples: Microsoft Windows, MS Office (commercial license)

### Linux — Introduction

- **Linux** = open-source OS kernel; many **distributions (distros)** package it with apps
- **Ubuntu** = popular beginner-friendly Linux distro with GUI (similar tasks to Windows)
- Used on servers worldwide; also runs on old PCs via dual-boot or live USB

### Windows vs Ubuntu (quick comparison)

| Task | Windows | Ubuntu (Linux) |
|---|---|---|
| File manager | File Explorer | Files (Nautilus) |
| App store | Microsoft Store | Ubuntu Software / Snap Store |
| Settings | Settings (Win+I) | Settings app |
| Terminal | Command Prompt / PowerShell | Terminal (bash) |
| Install apps | .exe / Microsoft Store | .deb / Snap / apt |
| Office suite | MS Office | LibreOffice (pre-installed) |

### Trying Ubuntu (lab option)

1. Download Ubuntu ISO from [ubuntu.com](ubuntu.com)
2. Create bootable USB (Rufus) **or** use virtual machine (VirtualBox)
3. Explore desktop, Files app, Firefox, LibreOffice, Terminal

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## 3. Desktop and Taskbar (Windows)

### Desktop elements

- **Icons:** Shortcuts to programs/files
- **Taskbar:** Bottom bar — Start, pinned apps, system tray, clock
- **System tray:** Network, volume, battery, date/time
- **Wallpaper:** Background image (right-click desktop → Personalize)

### Start Menu

- Search box — type app name
- Pinned apps — Word, Excel, Settings
- Power button — Shut down, Restart, Sleep

### Keyboard shortcuts (must know)

| Shortcut | Action |
|---|---|
| Win | Open Start menu |

| | |
|---|---|
| Win + E | File Explorer |
| Win + D | Show desktop |
| Win + L | Lock computer |
| Win + I | Settings |
| Alt + Tab | Switch between open windows |
| Alt + F4 | Close active window |
| Ctrl + C / V / X | Copy / Paste / Cut |
| Ctrl + Z | Undo |
| Ctrl + S | Save |
| F2 | Rename selected item |
| Delete | Send to Recycle Bin |
| Shift + Delete | Permanent delete |

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## 4. File Explorer

**Open:** Win + E or taskbar folder icon

### Parts of File Explorer

- **Navigation pane** (left): Quick access, This PC, drives
- **Address bar:** Current path e.g. `C:\Users\Student\Documents`
- **Ribbon:** Home, Share, View tabs
- **File list:** Name, Date modified, Type, Size

### Views

- **Icons / List / Details / Tiles** — View tab
- **Sort** by name, date, size, type
- **Search box** — find files by name inside folder

### Drives

- **C:** — Usually Windows and programs
- **D:** or others — Data, DVD, USB when connected

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## 5. File and Folder Operations

### Create folder

1. Open desired location in File Explorer
2. Right-click → New → Folder
3. Type name → Enter

### Copy / Move

- **Copy:** Ctrl+C → go to destination → Ctrl+V (original stays)
- **Move (Cut):** Ctrl+X → destination → Ctrl+V
- **Drag-drop:** Drag with left mouse; hold **Shift** to move, **Ctrl** to copy

### Rename

- Select item → F2 → type new name → Enter
- Do not change file extension (.docx) unless you know the type

### Delete and Restore

- Delete → Recycle Bin (can restore)
- Empty Recycle Bin → permanent delete
- Shift + Delete → skip Recycle Bin

### File naming rules (Windows)

- Invalid characters: `\ / : * ? " < > |`
- Extensions identify type: `report.docx`, `data.xlsx`
- Case usually not sensitive: `File.txt` = `file.txt`

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## 6. Installing and Removing Programs

### Installing software (Windows)

#### Method 1 — Installer (.exe / .msi)

1. Download from official website only (avoid unknown links)
2. Double-click installer → follow wizard (Next → I Agree → Install)
3. Choose install location if asked (default is usually fine)
4. Finish → launch from Start menu

#### Method 2 — Microsoft Store

1. Open Microsoft Store from Start
2. Search app → Get / Install
3. Updates often automatic

#### Method 3 — Open-source on Windows

- Download LibreOffice, Firefox, VLC from official sites — same wizard install

**Ubuntu equivalent:** Ubuntu Software → search → Install, or Terminal: `sudo apt install libreoffice`

### Removing / Uninstalling software (Windows)

#### Settings method (recommended)

1. Win + I → **Apps** → **Installed apps** (or Apps & features)
2. Search program name → ⋯ → **Uninstall**
3. Follow prompts; restart if asked

#### Control Panel method (classic)

1. Control Panel → Programs → Uninstall a program
2. Select program → Uninstall

**Ubuntu equivalent:** Ubuntu Software → Installed → Remove, or `sudo apt remove packagename`

### Safe installation habits

- Install only needed software; avoid pirated/cracked copies
- Uncheck bundled extra toolbars or unwanted apps during setup
- Keep Windows Update enabled for security patches
- Remove unused programs to free disk space

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## 7. Control Panel and Settings

**Settings (Win + I):** Modern — System, Devices, Network, Personalization, Accounts, Update

**Control Panel:** Classic — Programs, Devices and Printers, User Accounts

### Common tasks

| Task | Path |
|---|---|
| Change display | Settings → System → Display |
| Add printer | Settings → Bluetooth & devices → Printers |
| Uninstall program | Settings → Apps → Installed apps |
| Create user | Settings → Accounts → Family & other users |
| Date/time | Taskbar clock → Adjust date/time |
| Windows Update | Settings → Windows Update |

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## 8. Built-in Utilities

| Program | Use | Open |
|---|---|---|
| Notepad | Plain text | Start → Notepad |
| WordPad | Rich text | Start → WordPad |
| Calculator | Standard / Scientific | Start → Calculator |
| Paint | Simple drawing | Start → Paint |
| Snipping Tool | Screenshot | Start → Snipping Tool |
| Command Prompt | Text commands | Start → cmd |
| Task Manager | Running programs, performance | Ctrl+Shift+Esc |

### Task Manager uses

- End frozen program (End task)
- Check CPU/RAM usage (Performance tab)
- Startup programs (Startup tab)

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## 9. User Accounts and Security

- **Administrator:** Full control — install software, change settings
- **Standard user:** Limited — safer for daily use
- **Password / PIN:** Lock screen protection
- **Windows Update:** Keep OS patched — Settings → Windows Update

### Good habits

- Lock PC when away (Win + L)
- Do not share password
- Regular backup to USB or cloud
- Use antivirus (Windows Defender built-in)

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## 10. Compression and Shortcuts

### Zip (compress)

- Right-click folder → Send to → Compressed (zipped) folder
- Reduces size for email/upload

### Desktop shortcut

- Right-click program → Send to → Desktop (create shortcut)
- Or right-click desktop → New → Shortcut → browse to .exe

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## Lab Tasks

1. Create folder structure: `Documents\CCC\Unit2\Notes` and `Documents\CCC\Unit2\Labs`
2. Create `my-info.txt` in Notes with your name, course, date
3. Copy to Labs; rename copy to `my-info-backup.txt`
4. Practice shortcuts: Win+E, Win+D, Win+L, Alt+Tab, F2, Ctrl+C/V/X
5. Take screenshot with Snipping Tool; save as PNG in Labs
6. Open Task Manager — note CPU and memory usage %
7. Create zip of Labs folder; compare size with original folder
8. **Install:** Install LibreOffice or Firefox from official site (or use Store); note install steps
9. **Uninstall:** Remove a practice app via Settings → Apps → Uninstall (teacher-approved app only)
10. **Linux demo:** Boot Ubuntu live USB or VM — open Files, Firefox, and LibreOffice Writer; compare with Windows

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## Summary Checklist

- [ ] I can explain five functions of an OS
- [ ] I know open source vs closed source with examples

- ☐ I can name basic Ubuntu apps equivalent to Windows tools
- ☐ I can navigate File Explorer and manage files/folders
- ☐ I can install and uninstall a program safely
- ☐ I know 10+ keyboard shortcuts
- ☐ I can use Notepad, Calculator, Snipping Tool, Task Manager

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## Quick MCQs (10 Questions)

**1.** Which is an open-source operating system?

- a) Windows 11
- b) macOS
- c) Ubuntu Linux
- d) iOS

**2.** Primary function of an operating system is:

- a) Only playing games
- b) Managing hardware and providing user interface
- c) Replacing the CPU
- d) Storing data permanently without software

**3.** Shortcut to open File Explorer in Windows:

- a) Win + E
- b) Win + P
- c) Ctrl + E
- d) Alt + E

**4.** To move a file (not copy) using keyboard:

- a) Ctrl+C then Ctrl+V
- b) Ctrl+X then Ctrl+V
- c) Shift+Delete
- d) F2

**5.** Recommended way to uninstall a program in Windows 10/11:

- a) Delete folder from Program Files only
- b) Settings → Apps → Installed apps → Uninstall
- c) Rename the .exe file
- d) Empty Recycle Bin

**6.** LibreOffice is an example of:

- a) Hardware
- b) Open-source application suite
- c) Virus
- d) CPU type

**7.** Shift + Delete does what?

- a) Sends file to Recycle Bin
- b) Permanently deletes without Recycle Bin

- c) Renames file
- d) Copies file

**8.** Win + L is used to:

- a) Open LibreOffice
- b) Lock the computer
- c) Log off permanently
- d) Open Linux terminal

**9.** Ubuntu's graphical file manager is commonly called:

- a) File Explorer
- b) Finder
- c) Files (Nautilus)
- d) Notepad

**10.** Task Manager is opened by:

- a) Ctrl+Alt+Del only
- b) Ctrl+Shift+Esc
- c) Win+T
- d) F12

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### MCQ Answers

1-c | 2-b | 3-a | 4-b | 5-b | 6-b | 7-b | 8-b | 9-c | 10-b

# Module 3: Word Processing (12 Hours)

**Course:** CCC 90-Hour DCA | **Module duration:** 12 hours

## Learning Objectives

After this module you will be able to:

- Create, save, open, and print documents in MS Word and LibreOffice Writer
- Apply character and paragraph formatting
- Use bullets, numbering, borders, tables, and images
- Insert headers, footers, and page numbers
- Use find/replace, spell check, and styles
- Prepare professional **letters** and **resumes**
- Perform basic mail merge

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## 1. Introduction to Word Processing

A **word processor** creates and edits text documents with formatting, unlike plain Notepad.

| Software | Developer | Default extension | Open with |
|---|---|---|---|
| **MS Word** | Microsoft | `.docx` | Word, Writer (partial) |
| **LibreOffice Writer** | The Document Foundation (open source) | `.odt` | Writer, Word (partial) |

**CCC note:** Skills transfer between Word and Writer — menus differ slightly but concepts are the same.

### Open a blank document

- **Word:** Start → Microsoft Word → Blank document
- **Writer:** Start → LibreOffice Writer (or search Writer)

### Screen layout (both apps)

| Area | Purpose |
|---|---|
| Title bar | Document name, window controls |
| Ribbon / Menus | Home, Insert, Layout/Page, Review, View |
| Ruler | Margins, indents |
| Document area | Typing space |
| Status bar | Page count, word count, zoom |

### Save

- **Ctrl + S** — first time: choose location and filename
- **Save As** — new name, `.pdf` , or `.odt` / `.docx` format
- **Word:** File → Save As → PDF

- **Writer:** File → Export as PDF

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## 2. MS Word vs LibreOffice Writer — Feature Parity

| Task | MS Word | LibreOffice Writer |
|---|---|---|
| Bold / Italic | Ctrl+B / Ctrl+I | Same |
| Alignment | Home → alignment | Home (toolbar) → alignment |
| Line spacing | Home → Line spacing | Format → Paragraph → Line spacing |
| Table | Insert → Table | Insert → Table |
| Picture | Insert → Pictures | Insert → Image |
| Header/Footer | Insert → Header & Footer | Insert → Header/Footer |
| Page break | Ctrl+Enter | Ctrl+Enter |
| Find/Replace | Ctrl+F / Ctrl+H | Ctrl+F / Ctrl+H |
| Spell check | Review → Spelling (F7) | Tools → Spelling (F7) |
| Styles | Home → Styles | Styles sidebar (F11) |
| Mail merge | Mailings tab | Tools → Mail Merge Wizard |
| Templates | File → New → search | File → New → Templates |

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## 3. Basic Text Entry and Editing

- **Enter** — new paragraph | **Shift + Enter** — line break in same paragraph
- **Backspace** — delete left | **Delete** — delete right
- **Select:** drag mouse, Ctrl+A (all), Shift+arrows
- **Undo / Redo:** Ctrl+Z / Ctrl+Y (Writer: Ctrl+Shift+Z for redo)

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## 4. Character Formatting

| Tool | Shortcut | Use |
|---|---|---|
| Bold | Ctrl+B | Headings, emphasis |
| Italic | Ctrl+I | Titles, foreign words |
| Underline | Ctrl+U | Underline |
| Font size | — | 12 pt body; 14–16 headings |
| Font color / Highlight | — | Emphasis |
| Clear formatting | — | Reset to default |

**Fonts:** Arial, Times New Roman, Calibri (Word); Liberation Serif/Sans (Writer)

## 5. Paragraph Formatting

| Setting | Notes |
| --- | --- |
| Alignment | Left, Center, Right, **Justify** (formal letters, essays) |
| Line spacing | 1.0, 1.5, 2.0 — resumes often use 1.0–1.15 |
| Before/After spacing | Space between paragraphs |
| Indent | First-line indent for essays; hanging indent for references |
| Bullets / Numbering | Lists and outlines |

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## 6. Page Layout

- **Margins:** Layout (Word) / Format → Page (Writer) — Normal, Narrow, Custom
- **Orientation:** Portrait (letters, resumes) / Landscape (tables, certificates)
- **Size:** A4 (standard India)
- **Columns:** Layout → Columns — newsletters
- **Page Break:** Ctrl+Enter

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## 7. Insert — Tables, Pictures, Headers

### Table

Insert → Table → choose rows×columns
**Tab** = next cell | Table design: borders, shading, header row

### Pictures

Insert → Picture/Image → resize; **Wrap Text** (Square, Tight) for text flow

### Header, Footer, Page Number

Insert → Header/Footer → type institute name or "Resume"
Insert → Page Number → bottom center (avoid on one-page resume)

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## 8. Find, Replace, Spell Check

- **Find:** Ctrl+F | **Replace:** Ctrl+H
- **Spelling:** F7 or Review/Tools → Spelling
- **Word count:** Review → Word Count (Writer: Tools → Word Count)

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## 9. Styles and Templates

**Styles:** Heading 1, Heading 2, Normal — consistent structure for reports

**Built-in templates**

- **Word:** File → New → search "Resume", "Cover letter", "Formal letter"
- **Writer:** File → New → Templates → Letters, CV/resume templates

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## 10. Practical: Formal Letter

### Structure of a formal letter

```
[Your address – top right or left]
[Date]

[Recipient name & designation]
[Organization address]

Subject: [Bold, one line]

Respected Sir/Madam,

[Body – 2–3 paragraphs, justified]

Yours faithfully,
[Your name]
[Contact: phone, email]
```

### Formatting checklist for letters

- A4, margins 2.5 cm (Normal)
- Font: Times New Roman 12 pt or Arial 11 pt
- Subject line bold; paragraphs justified
- Leave space for signature above your name

### Lab reference letter topic

Write to the Principal requesting **study leave** or **fee concession** — one page, proper salutation and subject.

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## 11. Practical: Resume (CV)

### Resume sections (one page for freshers)

1. **Name & contact** — phone, email, city (no need for full address in short CV)
2. **Career objective** — 2 lines (optional for CCC level)
3. **Education** — Class, board, year, percentage (table or lines)
4. **Skills** — MS Word, Excel, typing speed, English/Hindi
5. **Certifications** — CCC, NCVRT, etc.
6. **Declaration** — optional one line

### Resume formatting tips

- **One page** for students; clean font (Calibri 11, Arial 10)

- Use **bold** for section headings; avoid colours overload
- **No photo** unless employer asks (India government forms sometimes require)
- Export **PDF** before submitting — layout stays fixed
- Word: File → New → "Simple resume" | Writer: Template → Curriculum vitae

---

## 12. Mail Merge (Overview)

**Purpose:** One letter to many people with personalized name/address.

### Steps (Word)

1. Mailings → Start Mail Merge → Letters
2. Select Recipients → use Excel list or type new list
3. Insert Merge Fields (Name, Address, City)
4. Preview Results → Finish & Merge → Print or Edit documents

### Writer equivalent

Tools → Mail Merge Wizard → follow steps (data source: spreadsheet .ods/.xlsx)

---

## 13. Print and Export

- **Print:** Ctrl+P — printer, pages, copies
- **PDF:** Save As / Export as PDF — submit assignments and resumes

---

## Lab Tasks

1. Type a 1-page essay "Uses of Computer" — Heading 1 title, justify, 1.5 spacing
2. Create **formal letter** to Principal (study leave) — correct layout on A4
3. Create **one-page resume** using a template; fill your real education and skills; export PDF
4. Build 4×4 marks table (Name, Roll, Subject, Marks)
5. Insert image with Square text wrap; add header "NITA Classes" and page numbers
6. Find & replace one word throughout a document
7. Repeat letter OR resume in **LibreOffice Writer** — save as `.odt` and export PDF
8. Optional: Mail merge letter to 5 recipients using small Excel/Calc list

---

## Summary Checklist

- [ ] I can work in both Word and Writer for basic tasks
- [ ] I can format text and paragraphs professionally
- [ ] I can create a formal letter with subject and salutation
- [ ] I can create a one-page resume and export PDF
- [ ] I can insert tables, images, header/footer
- [ ] I can run spell check and find/replace
- [ ] I understand basic mail merge steps

---

## Quick MCQs (10 Questions)

**1.** Default LibreOffice Writer file extension is:

- a) .docx
- b) .odt
- c) .xlsx
- d) .pptx

**2.** Shortcut for Save in Word/Writer:

- a) Ctrl+P
- b) Ctrl+S
- c) Ctrl+N
- d) F7

**3.** For a formal letter in India, paper size is usually:

- a) Letter
- b) A4
- c) Legal only
- d) A3

**4.** Mail merge is used to:

- a) Merge two printers
- b) Create personalized bulk letters from one template
- c) Delete duplicate pages
- d) Compress files

**5.** In Writer, spell check is commonly under:

- a) Tools → Spelling
- b) Insert → Picture
- c) View → Zoom
- d) File → Print

**6.** Page break shortcut:

- a) Ctrl+Enter
- b) Shift+Enter
- c) Alt+Enter
- d) Ctrl+Break

**7.** A fresher resume should ideally be:

- a) 5 pages with graphics
- b) One page, clear sections, PDF for submission
- c) Only handwritten
- d) Landscape orientation only

**8.** Justify alignment means:

- a) Text aligned left only
- b) Both left and right edges straight
- c) Text in centre

- d) Vertical text

**9.** Which is open-source word processor?

- a) MS Word
- b) LibreOffice Writer
- c) Notepad
- d) Paint

**10.** Header and footer appear:

- a) Only on first page always
- b) At top and bottom of pages (can vary by section)
- c) Only in PDF
- d) Behind the text always

---

### MCQ Answers

1-b | 2-b | 3-b | 4-b | 5-a | 6-a | 7-b | 8-b | 9-b | 10-b

# Module 4: Spreadsheet (12 Hours)

**Course:** CCC 90-Hour DCA | **Module duration:** 12 hours

## Learning Objectives

After this module you will be able to:

- Navigate workbook, sheets, cells, and ranges in Excel and LibreOffice Calc
- Enter and format data (numbers, text, dates, currency)
- Write formulas and use common functions (SUM, AVERAGE, IF, COUNTIF)
- Apply relative and absolute cell references
- Sort, filter, and use conditional formatting
- Create and customize charts
- Set print area, headers, and page setup for worksheets

**Focus:** Spreadsheet skills in **MS Excel** and **LibreOffice Calc** — not database tools (Access/Base).

---

## 1. Introduction to Spreadsheets

A **spreadsheet** organizes data in rows and columns for calculation, analysis, and charts.

| Software | Extension | Open source? |
|---|---|---|
| **MS Excel** | `.xlsx` | No (Microsoft) |
| **LibreOffice Calc** | `.ods` | Yes |

### Basic terms

- **Workbook** = entire file
- **Worksheet (Sheet)** = tab (Sheet1, Sheet2...)
- **Cell** = intersection of row and column
- **Cell address:** Column letter + row number — **A1**, **D15**
- **Range:** A1:C10 (rectangle)

### Fill handle

Drag small square at cell corner to copy formula or extend series (1,2,3... or Jan, Feb...)

---

## 2. MS Excel vs LibreOffice Calc — Parity

| Task | MS Excel | LibreOffice Calc |
|---|---|---|
| New workbook | Ctrl+N | Ctrl+N |
| Save | Ctrl+S | Ctrl+S |
| Format cells | Ctrl+1 | Ctrl+1 |
| Sum formula | `=SUM(A1:A10)` | Same syntax |

| | | |
|---|---|---|
| AutoSum | Home → AutoSum | Sigma Σ on toolbar |
| Sort/Filter | Data tab | Data menu |
| Chart | Insert → Chart | Insert → Chart |
| Sheet tab | Bottom tabs | Bottom tabs |
| Print preview | Ctrl+P | Ctrl+P |
| Export PDF | File → Save As PDF | File → Export as PDF |

**Formula syntax is nearly identical** — CCC exams often use Excel notation; it works in Calc.

---

## 3. Data Entry and Formatting

### Data types

- **Text** — left-aligned (names, roll numbers as text)
- **Numbers** — right-aligned (marks, amounts)
- **Dates** — format dd-mm-yyyy (India)

### Home tab / Formatting toolbar

| Tool | Use |
|---|---|
| Number format | General, Number, Currency ₹, Percentage, Date |
| Decimal places | Increase/decrease |
| Bold / Borders | Table appearance |
| Merge & Center | Title across columns |
| Wrap Text | Multi-line in one cell |

**Format Cells (Ctrl+1):** Number, Alignment, Font, Border, Fill

---

## 4. Formulas (always start with `=`)

| Formula | Example | Result |
|---|---|---|
| Addition | `=A1+B1` | Sum of two cells |
| Subtraction | `=A1-B1` | Difference |
| Multiplication | `=A1*B1` | Product |
| Division | `=A1/B1` | Quotient |
| Average | `=AVERAGE(A1:A10)` | Mean |
| Sum | `=SUM(A1:A10)` | Total |

| | | |
|---|---|---|
| Count | `=COUNT(A1:A10)` | Count numbers |
| Max / Min | `=MAX(A1:A10)` / `=MIN()` | Highest / lowest |
| Percentage | `=B1/C1*100` | % marks |

### Cell references

- **Relative:** `=B2*C2` — changes when copied down (B3*C3, B4*C4...)
- **Absolute:** `=$B$1` — fixed when copying (press **F4** in Excel/Calc)
- **Mixed:** `=$B1` or `=B$1` — lock one part only

**Use case:** Tax rate in cell B1 ($B$1) applied to every row of amounts.

---

## 5. Important Functions (Exam Focus)

| Function | Syntax | Use |
|---|---|---|
| SUM | `=SUM(B2:B20)` | Total |
| AVERAGE | `=AVERAGE(B2:B20)` | Average |
| COUNT | `=COUNT(B2:B20)` | Count numeric cells |
| COUNTA | `=COUNTA(A2:A20)` | Count non-empty cells |
| MAX / MIN | `=MAX()`, `=MIN()` | Extremes |
| IF | `=IF(C2>=40,"Pass","Fail")` | Condition |
| COUNTIF | `=COUNTIF(B2:B20,">=40")` | Count if condition |
| SUMIF | `=SUMIF(A2:A20,"Delhi",B2:B20)` | Sum matching criteria |
| CONCAT / & | `=A2&" "&B2` | Join text (Calc: CONCATENATE) |
| TODAY | `=TODAY()` | Current date |
| ROUND | `=ROUND(A2,2)` | Round decimals |

**Nested IF (grades):**
`=IF(B2>=60,"First",IF(B2>=45,"Second","Fail"))`

---

## 6. Sort and Filter

- **Sort:** Data → Sort A to Z / Z to A / Custom sort (multiple columns)
- **Filter:** Data → Filter — dropdown arrows in headers; show only matching rows
- **Remove Duplicates:** Data → Remove Duplicates (Excel); Calc: Data → More filters → Standard filter

**Example:** Sort students by total marks descending; filter total > 200.

---

## 7. Conditional Formatting

Home → Conditional Formatting (Calc: Format → Conditional)

- Highlight Cell Rules — greater than, less than, between
- Color scales — heat map
- Data bars — visual bars in cells
- Icon sets — arrows, traffic lights

**Example:** Marks < 40 → red fill; marks > 75 → green fill.

---

## 8. Charts

Insert → Chart → select data range with labels

| Chart type | Best for |
|---|---|
| Column / Bar | Compare categories (subject-wise marks) |
| Line | Trends over time (monthly sales) |
| Pie | Parts of whole (pass/fail proportion) |
| Area | Volume over time |

**Steps:** Select data including row/column labels → Insert chart → add title, legend, axis labels.

---

## 9. Page Setup and Print

- **Print area:** Page Layout → Print Area → Set Print Area
- **Print titles:** Repeat row 1 (headers) on every printed page
- **Orientation:** Portrait (tall data) / Landscape (wide tables)
- **Margins:** Narrow for more columns on one page
- **Gridlines:** Print gridlines option for clarity
- **Preview:** Ctrl+P

---

## 10. Multiple Sheets

- **Add sheet:** + tab or Shift+F11
- **Rename:** Double-click tab
- **3D formula:** `=SUM(Sheet1:Sheet3!B2)` — same cell across sheets
- **Link:** `=Sheet2!A1` — reference another sheet

---

## 11. Typical CCC Spreadsheet Tasks

| Task | Approach |
|---|---|
| Student marksheet | Formulas: Total, Average, %, Grade (IF) |

| | |
|---|---|
| Fee collection | SUM, date column, receipt number |
| Attendance % | COUNTA, division, percentage format |
| Inventory list | Sort by quantity; filter low stock |
| Simple budget | SUM by category; pie chart of expenses |

---

## Lab Tasks

1. Create marks sheet: 10 students, 3 subjects — Total, Average, % (formulas)
2. Add **Pass/Fail** column with `=IF(total>=120,"Pass","Fail")` or per-subject IF
3. Use COUNTIF to count passes (>=40) in each subject
4. Sort by total descending; filter students with total > 200
5. Conditional format: fail marks red, distinction (>75) green
6. Column chart of totals; pie chart of pass/fail count
7. Set print area and repeat header row; print preview
8. Use **absolute reference:** tax rate in $B$1 applied to all amount rows
9. Repeat tasks 1–4 in **LibreOffice Calc** — save as `.ods` and export PDF

---

## Summary Checklist

- [ ] I can work in Excel and Calc with same formulas
- [ ] I write SUM, AVERAGE, IF, COUNTIF correctly
- [ ] I use relative and absolute references (F4)
- [ ] I sort, filter, and apply conditional formatting
- [ ] I create column and pie charts
- [ ] I set print titles and print area
- [ ] I know spreadsheet ≠ database (no Access focus in this module)

---

## Quick MCQs (10 Questions)

**1.** Cell at column D row 10 is:

- a) 10D
- b) D10
- c) D:10
- d) 4J

**2.** Every formula in Excel/Calc must begin with:

- a) +
- b) =
- c) #
- d) @

**3.** To keep cell B1 fixed when copying a formula down, use:

- a) B1
- b) $B$1

- c) B$1 only in Excel Mac
- d) #B1

**4.** Function to count only numeric cells in a range:

- a) COUNTA
- b) COUNT
- c) SUM
- d) LEN

**5.** LibreOffice Calc default file extension:

- a) .xlsx
- b) .ods
- c) .odt
- d) .mdb

**6.** `=IF(C2>=40,"Pass","Fail")` returns "Fail" when:

- a) C2 is 50
- b) C2 is 40
- c) C2 is 35
- d) C2 is text "Pass"

**7.** Best chart to show marks of 5 subjects for one class:

- a) Pie chart of one student
- b) Column/Bar chart
- c) Scatter with no labels
- d) Only line chart for categories

**8.** Conditional formatting is used to:

- a) Change hardware
- b) Automatically format cells based on rules
- c) Install fonts
- d) Merge sheets

**9.** Shortcut to open Format Cells dialog:

- a) Ctrl+1
- b) Ctrl+F
- c) Alt+Tab
- d) F5

**10.** A workbook contains:

- a) Only one cell
- b) One or more worksheets
- c) Only charts
- d) Only macros

---

### MCQ Answers

1-b | 2-b | 3-b | 4-b | 5-b | 6-c | 7-b | 8-b | 9-a | 10-b

# Module 5: Presentation (8 Hours)

**Course:** CCC 90-Hour DCA | **Module duration:** 8 hours

## Learning Objectives

After this module you will be able to:

- Create multi-slide presentations in MS PowerPoint and LibreOffice Impress
- Apply themes, layouts, and consistent design
- Add and format text, bullets, images, charts, and tables
- Use slide transitions and object animations appropriately
- Run slideshow with keyboard navigation and presenter tools
- Export presentations as PDF, video, or slideshow file

---

## 1. Introduction to Presentation Software

A **presentation** = sequence of **slides** shown to an audience — classes, seminars, project demos.

| Software | Extension | Notes |
|---|---|---|
| **MS PowerPoint** | `.pptx` | Industry standard in offices |
| **LibreOffice Impress** | `.odp` | Open source; free with Ubuntu/LibreOffice |

### Open blank presentation

- **PowerPoint:** Start → PowerPoint → Blank Presentation
- **Impress:** Start → LibreOffice Impress

### Key terms

- **Slide** = one screen of content
- **Slide layout** = arrangement of placeholders (title, content, blank)
- **Theme** = coordinated colours, fonts, backgrounds
- **Transition** = effect **between** slides
- **Animation** = effect **on objects within** a slide

---

## 2. MS PowerPoint vs LibreOffice Impress — Parity

| Task | PowerPoint | Impress |
|---|---|---|
| New slide | Ctrl+M / Home → New Slide | Slide → New Slide |
| Slide show from start | F5 | F5 |
| From current slide | Shift+F5 | Shift+F5 |
| Duplicate slide | Right-click → Duplicate | Slide → Duplicate slide |
| Insert picture | Insert → Pictures | Insert → Image |

| | | |
|---|---|---|
| Transitions | Transitions tab | Slide Transition sidebar |
| Animations | Animations tab | Custom Animation |
| Export PDF | File → Save As PDF | File → Export as PDF |
| Video export | File → Export → Video | File → Export (varies by version) |
| Direct slideshow file | .ppsx | Configure in slide show settings |

---

## 3. Views

| View | PowerPoint | Use |
|---|---|---|
| Normal | Default | Edit slides |
| Slide Sorter | View → Slide Sorter | Reorder many slides |
| Reading View | View → Reading View | Preview full screen |
| Slide Show | F5 | Present to audience |

**Impress:** Similar — Normal, Slide Sorter, Slide Show (bottom tabs or View menu).

---

## 4. Creating and Managing Slides

### New slide

- Home → New Slide (Ctrl+M)
- Choose **layout:** Title Slide, Title and Content, Two Content, Blank, Comparison

### Slide operations

- **Duplicate:** Right-click thumbnail → Duplicate
- **Delete:** Select slide → Delete key
- **Reorder:** Drag in left pane or Slide Sorter view

### Text and lists

- **Title placeholder** — main heading (one per slide)
- **Content placeholder** — bullets and sub-bullets
- **Tab / Shift+Tab** — increase/decrease list level

**6×6 rule:** About 6 bullets max, ~6 words each — keeps slides readable.

---

## 5. Design and Themes

### PowerPoint

- **Design tab** → **Themes** — colours, fonts, effects
- **Variants** — colour variations
- **Design Ideas** (Microsoft 365) — AI layout suggestions
- **Slide Size:** Standard (4:3) or Widescreen (16:9)

### Impress

- **Slide** → **Apply Layout** / **Master Slide** for consistent branding
- **Format** → **Slide Design** or sidebar for themes

### Background

- Design → Format Background — solid colour, gradient, or picture
- Keep contrast high — dark text on light background (or reverse)

---

## 6. Insert Objects

| Object | Use on slide |
| --- | --- |
| Pictures | Photos, logos — resize, crop |
| Shapes | Arrows, boxes for diagrams |
| Icons / SmartArt | Process flows, org charts (PowerPoint SmartArt) |
| Chart | Data from Excel/Calc |
| Table | Small tabular data |
| Video / Audio | Demos, background music (use sparingly) |

**Impress:** Insert menu mirrors most options; charts link to Calc data.

---

## 7. Transitions (Between Slides)

**PowerPoint:** Transitions tab | **Impress:** Slide Transition deck

- Apply to one slide or **Apply To All**
- Set **duration** (speed)
- **Advance:** On mouse click or after X seconds

**CCC tip:** Use simple effects (Fade, Push, Wipe) — avoid distracting spins in formal talks.

---

## 8. Animations (On-Slide Objects)

**PowerPoint:** Animations tab | **Impress:** Custom Animation

| Type | Example |
| --- | --- |
| Entrance | Appear, Fly In, Fade |
| Emphasis | Pulse, Grow/Shrink |
| Exit | Disappear |
| Motion path | Move along path |

**Start options:** On Click / With Previous / After Previous
**Animation Pane:** Control order of multiple effects

**Example:** Bullet list — appear one by one on click.

---

## 9. Slide Show Delivery

| Action | Key / command |
|---|---|
| Start from beginning | F5 |
| Start from current slide | Shift+F5 |
| Next slide | Click, Space, Enter, → |
| Previous slide | ← or Backspace |
| Black screen | B |
| End show | Esc |
| Pen (draw on slide) | Ctrl+P (PowerPoint) |

**Presenter View (PowerPoint):** Slide Show → Use Presenter View — notes, timer, next slide preview on your screen only.

---

## 10. Print and Export

- **Print:** Full slides, **handouts** (2 or 6 per page), notes pages
- **PDF:** File → Save As / Export as PDF — sharing and printing
- **Video:** File → Export → Create Video (.mp4) — PowerPoint
- **Slideshow file:** `.ppsx` — opens directly in slide show mode (PowerPoint)

---

## 11. Sample Presentation Outline (CCC Topic)

**Title:** Digital India / Computer Literacy / Cyber Awareness

| Slide # | Content |
|---|---|
| 1 | Title slide — topic, your name, institute |
| 2 | Introduction — what is the topic? |
| 3–4 | Key points (bullets, one idea per slide) |
| 5 | Chart or SmartArt — statistics or process |
| 6 | Image slide — relevant photo with caption |
| 7 | Summary — 3–4 takeaway points |
| 8 | Thank you — contact / questions |

---

## Lab Tasks

1. Create **8-slide** presentation: "Digital India" (or teacher-assigned topic)
2. Apply one theme consistently in PowerPoint; repeat in Impress with equivalent theme
3. Use Title + Content layouts; no overcrowded slides (6×6 rule)
4. Add **transitions** to all slides (same simple effect)
5. Animate bullet list — appear one by one on click
6. Insert picture and one chart or SmartArt/diagram
7. Run slideshow with F5; practice B (black screen) and Esc
8. Export as **PDF** from both PowerPoint and Impress
9. Optional: Export short video or save `.ppsx` slideshow file

---

## Summary Checklist

- [ ] I can create slides in PowerPoint and Impress
- [ ] I apply themes and consistent layouts
- [ ] I know transition vs animation difference
- [ ] I insert images, charts, and shapes
- [ ] I present with F5 and keyboard navigation
- [ ] I export PDF (and optional video/ppsx)

---

## Quick MCQs (10 Questions)

**1.** LibreOffice presentation software is:

- a) Writer
- b) Calc
- c) Impress
- d) Draw only

**2.** Shortcut to start slideshow from beginning:

- a) F1
- b) F5
- c) Ctrl+S
- d) Alt+F5

**3.** Transition applies to:

- a) Text colour inside a shape
- b) Movement between slides
- c) Spell check
- d) Printer setup

**4.** Animation applies to:

- a) Objects on a slide
- b) Only the first slide
- c) File name
- d) Operating system

**5.** Default PowerPoint file extension:

- a) .docx
- b) .pptx
- c) .xlsx
- d) .odp

**6.** Impress default file extension:

- a) .pptx
- b) .odp
- c) .ods
- d) .pdf

**7.** 6×6 guideline suggests:

- a) 6 slides only per presentation
- b) Roughly 6 bullets with short text per slide
- c) 6 fonts per slide
- d) 6 hours per slide

**8.** Shift+F5 starts slideshow:

- a) From first slide always
- b) From current slide
- c) In black screen only
- d) Without keyboard

**9.** Handout printing means:

- a) Multiple small slides per printed page
- b) Printing only animations
- c) Printing BIOS
- d) One slide per workbook

**10.** .ppsx file type:

- a) Word document
- b) Opens PowerPoint directly in slideshow mode
- c) Excel macro
- d) Linux kernel

---

### MCQ Answers

1-c | 2-b | 3-b | 4-a | 5-b | 6-b | 7-b | 8-b | 9-a | 10-b

# Subject 6: Internet & Web Browsing (8 Hours)

## Learning Objectives

- Explain internet, WWW, browser, URL, and ISP
- Connect to the internet using Wi-Fi and mobile data
- Browse websites safely using HTTPS and bookmarks
- Use search engines effectively for CCC exam topics
- Download and upload files correctly
- Understand cookies, cache, and browser privacy settings
- Follow safe browsing habits to avoid scams and malware

---

## 1. Internet Basics

**Internet** = a global network (vishwa jaal) of millions of computers connected through cables, fibre, and wireless links.

**WWW (World Wide Web)** = a system of linked web pages on the internet, accessed through browsers using HTTP/HTTPS.

| Term | Meaning |
|---|---|
| Website | Collection of related web pages on one server |
| Web page | Single HTML document shown in browser |
| Home page | First/main page of a website |
| Hyperlink | Clickable link to another page or site |
| Server | Computer that stores and sends web pages |
| Client | Your computer or phone that requests pages |

### How you connect to the internet

1. **ISP (Internet Service Provider)** — company that gives internet (Jio, Airtel, BSNL, local broadband)
2. **Modem** — converts signal from ISP line to digital data
3. **Router** — shares connection to many devices (often Wi-Fi)
4. **Wi-Fi** — wireless connection within home/office
5. **Mobile data (4G/5G)** — internet through SIM on phone

**IP Address** — unique number of a device on the internet (like house number).
**Domain name** — easy name like `google.com` instead of remembering IP numbers.
**DNS (Domain Name System)** — converts domain name to IP address.

---

## 2. Web Browser

A **browser** is software to open and view websites.

### Popular browsers (CCC exam)

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari (Apple devices)

### Parts of browser window

| Part | Use |
|---|---|
| Address bar | Type URL or search keywords |
| Tabs | Open many pages (Ctrl+T = new tab) |
| Back / Forward | Visit previous/next page |
| Refresh (F5) | Reload current page |
| Bookmarks / Favorites | Save site for quick access (Ctrl+D) |
| History (Ctrl+H) | List of pages you visited |
| Downloads (Ctrl+J) | Files saved from internet |
| Settings | Privacy, homepage, default search engine |

### Browser settings students should know

- Set **homepage** (page that opens on start)
- Clear **cache** and **cookies** when site behaves wrongly
- Enable **pop-up blocker** for safer browsing
- Turn on **safe browsing** warnings (Chrome/Edge)

---

## 3. URL — Uniform Resource Locator

**URL** = full web address of a page or file.

### Structure of a URL

```
https://www.example.gov.in:443/courses/index.html?page=1#top
└─┬─┘   └───────┬─────────┘ └┬┘ └───────┬───────┘ └───┬───┘ └┬┘
protocol     domain     port    path           query    fragment
```

| Part | Example | Meaning |
|---|---|---|
| Protocol | `https://` | Rules for connection |
| Domain | `www.example.gov.in` | Website name |
| Path | `/courses/index.html` | Location of page on server |
| Query | `?page=1` | Extra parameters |

### HTTP vs HTTPS

| Protocol | Security | Use |
|---|---|---|
| HTTP | Not encrypted | General reading only |
| HTTPS | Encrypted (SSL/TLS) | Login, banking, payments |

**HTTPS** shows a **padlock icon** in address bar. Always use HTTPS for passwords and online payment.

### Common top-level domains (TLD)

- `.com` — commercial
- `.org` — organizations
- `.edu` — educational
- `.gov.in` — Indian government (official)
- `.nic.in` — National Informatics Centre (government)

---

## 4. ISP and Connection Types

**ISP** provides internet access for a fee (monthly plan / prepaid recharge).

| Connection type | Speed | Typical use |
|---|---|---|
| Broadband (DSL/Fibre) | High | Home, office |
| Mobile data | Medium–high | Phone, hotspot |
| Dial-up (old) | Very slow | Rare today |

### Data usage terms

- **Bandwidth** — amount of data that can flow per second (Mbps)
- **Data limit** — GB allowed per month in mobile plan
- **Hotspot** — share phone internet to laptop via Wi-Fi

---

## 5. Search Engines

**Search engine** finds web pages matching your keywords.

### Popular search engines

Google, Bing, DuckDuckGo, Yahoo

### Effective search tips (CCC useful)

- Use **specific keywords**: `NIELIT CCC syllabus 2025` not just `computer`
- **Quotes** for exact phrase: `"Digital India mission"`
- **site:** filter: `site:gov.in UMANG app`
- **filetype:** filter: `filetype:pdf CCC study material`

- Open results from **trusted sources** (.gov.in, .edu, official institute sites)

### Evaluating search results

- Check **date** of article — old news may be wrong
- Avoid clickbait titles
- Cross-check important facts on official website

---

## 6. Download and Upload

| Action | Direction | Example |
|---|---|---|
| Download | Internet → your device | Save PDF, image, app |
| Upload | Your device → internet | Attach file to email, cloud upload |

### Safe download steps

1. Download only from **official or trusted** websites
2. Check file **extension** (.pdf, .docx, .jpg) — avoid unknown .exe unless sure
3. Scan with **antivirus** after download
4. Note **save location** (Downloads folder)
5. Do not download **pirated software** or cracked games (virus risk)

### Upload uses

- Email attachments
- Google Drive / OneDrive
- Online form document upload (admission, job application)

---

## 7. Cookies, Cache, and Sessions

### Cookies

Small text files stored by browser on your computer. Websites use cookies to:

- Remember login (session cookie)
- Save preferences (language, theme)
- Track usage (advertising cookies)

**Risk:** Third-party cookies can track browsing. Clear cookies or use private/incognito mode on shared PC.

### Cache

Temporary copies of images and pages stored locally for **faster loading**.
If a website shows old content, clear cache and refresh.

### Incognito / Private browsing

- Does not save history on local PC (still visible to ISP/school network admin)
- Useful on **cyber café** or library computer
- Always **log out** after banking/email on shared PC

---

## 8. Internet Safety While Browsing

### Common online risks

| Risk | What happens | Protection |
|---|---|---|
| Phishing site | Fake login page steals password | Check URL, padlock, official domain |
| Malware download | Virus in fake software | Official app stores only |
| Fake offers | "Free iPhone" scam links | Do not click suspicious ads |
| Public Wi-Fi sniffing | Data intercepted | Avoid banking on open Wi-Fi |

### Safe browsing checklist

- Use **HTTPS** for sensitive sites
- Keep **browser and Windows updated**
- Install **antivirus** and enable firewall
- Do not share **OTP, PIN, password** on call or chat
- Verify **.gov.in** for government schemes
- Report fraud at **cybercrime.gov.in**

### Parental and student discipline

- Limit screen time (samay prabandhan)
- Do not meet strangers from chat without guardian
- Think before posting personal photos (digital footprint)

---

## 9. Important CCC Exam Terms

| Term | Full form / meaning |
|---|---|
| WWW | World Wide Web |
| URL | Uniform Resource Locator |
| ISP | Internet Service Provider |
| HTTP | HyperText Transfer Protocol |
| HTTPS | HTTP Secure |
| DNS | Domain Name System |
| IP | Internet Protocol address |
| Wi-Fi | Wireless Fidelity |
| HTML | HyperText Markup Language (web page language) |

---

## Lab Tasks

1. **Connect:** Connect laptop/phone to Wi-Fi or mobile hotspot; note ISP name and signal strength.

2. **Browser tour:** Open Chrome or Edge — identify address bar, tabs, bookmarks, history, downloads.
3. **URL practice:** Visit `https://www.digitalindia.gov.in` — note protocol, domain, padlock icon.
4. **Search:** Search `NIELIT CCC exam pattern` — open one official/trusted result; bookmark it (Ctrl+D).
5. **Advanced search:** Use `site:gov.in DigiLocker` and open official page.
6. **Download:** Download a PDF from a government site; save to `Documents\CCC` folder.
7. **Upload:** Upload the same PDF to Google Drive (or institute cloud); note share link steps.
8. **Privacy:** Open browser settings — find option to clear cookies and cache; list 3 cookie uses.
9. **HTTPS check:** Compare HTTP vs HTTPS site — observe padlock difference.
10. **Safety list:** Write 5 signs of a fake/phishing website.

---

## Summary Checklist

- [ ] I can explain internet, WWW, browser, URL, and ISP
- [ ] I know parts of browser and shortcuts (Ctrl+T, Ctrl+D, Ctrl+H, Ctrl+J)
- [ ] I can read URL parts and difference between HTTP and HTTPS
- [ ] I can search effectively using keywords and site: filter
- [ ] I can download and upload files safely
- [ ] I understand cookies, cache, and private browsing
- [ ] I follow safe browsing rules for CCC and daily life

---

## Quick-Check MCQs (Questions Only)

1. Internet is best described as (a) a global network of connected computers (b) a single supercomputer in Delhi (c) only email system (d) a type of printer

2. WWW stands for (a) World Wide Web (b) World Windows Work (c) Wide Web Wire (d) Web World Write

3. Software used to open websites is called (a) browser (b) spreadsheet (c) compiler (d) scanner driver

4. URL is (a) the address of a web page (b) a type of RAM (c) printer command (d) keyboard layout

5. ISP provides (a) internet connection service (b) free MS Office license (c) CPU replacement (d) monitor repair

6. HTTPS compared to HTTP is (a) more secure with encryption (b) slower and never used (c) only for games (d) without any difference

7. Download means (a) saving a file from internet to your device (b) sending file from device to internet (c) deleting browser history (d) shutting down the PC

8. Cookies in a browser are (a) small files websites store on your computer (b) physical biscuits for lab (c) antivirus programs (d) types of CPU

9. To search only government India sites, useful operator is (a) site:gov.in (b) print:gov (c) delete:all (d) virus:scan

10. Padlock icon in address bar usually indicates (a) secure HTTPS connection (b) website is always free (c) no images on page (d) computer has virus

# Subject 7: Communication & Collaboration (8 Hours)

## Learning Objectives

- Create and use email for study and official communication
- Understand To, CC, BCC, subject, body, and attachments
- Use Gmail and basic Outlook webmail features
- Join and participate in online meetings (Google Meet, Zoom)
- Store and share files using cloud Drive services
- Understand social media basics and responsible use
- Follow email and online **netiquette** (sahi vyavahar)

---

## 1. Electronic Mail (Email)

**Email** = sending messages over the internet. Fast, cheap, and used for study, jobs, and government services.

### Email address format

```
username@domain.com
   |        |
   |        └── Domain (Gmail, Outlook, company, college)
   └── Your chosen name (unique on that domain)
```

**Examples:** `student@gmail.com`, `name@college.edu.in`

### Webmail vs Email client

| Type | Access | Examples |
|---|---|---|
| Webmail | Browser | Gmail.com, Outlook.com |
| Email client | Desktop/mobile app | Outlook app, Thunderbird |

**CCC focus:** Webmail (Gmail) is most common in exam and daily use.

---

## 2. Email Components

| Field | Purpose |
|---|---|
| **To** | Main recipient(s) — must have at least one |
| **CC (Carbon Copy)** | Extra recipients — everyone can see all CC names |
| **BCC (Blind Carbon Copy)** | Hidden copy — other recipients cannot see BCC list |
| **Subject** | Short title — must be clear and relevant |
| **Body** | Main message text |

| | |
|---|---|
| **Attachment** | File sent with email (PDF, photo, Word) |
| **Signature** | Name, phone, designation at bottom (optional) |

### When to use CC vs BCC

- **CC:** Inform teacher and class rep together — transparency
- **BCC:** Send newsletter to many people without exposing everyone's email (privacy)

### Attachment rules

- Click **paperclip icon** to attach file
- Gmail limit ~**25 MB** per email (large files → use Drive link)
- Scan attachments with antivirus if from unknown sender
- Common formats: .pdf, .docx, .jpg, .png

---

## 3. Gmail — Practical Guide

**Gmail** = Google's free email service (Google account required).

### Creating Gmail account

1. Visit gmail.com → Create account
2. Enter name, choose username, strong password
3. Add recovery phone/email for account recovery
4. Enable **2-Step Verification** (recommended)

### Main Gmail actions

| Action | Steps |
|---|---|
| Compose | Click Compose → fill To, Subject, Body → Send |
| Reply | Answer only the sender |
| Reply All | Answer sender + all To/CC recipients |
| Forward | Send received mail to another person |
| Star / Mark important | Flag mail for follow-up |
| Archive | Remove from Inbox but keep mail |
| Trash / Delete | Move to bin |
| Search | Use search bar — `from:teacher subject:CCC` |
| Labels / Folders | Organize mail by topic |

### Gmail safety settings

- Check **Last account activity** (bottom of Gmail) on shared PC
- Review **Filters** for auto-sorting institute mail
- Never share Gmail password; log out on cyber café

---

## 4. Email Writing and Netiquette

**Netiquette** = polite and correct behaviour on the internet (Internet + etiquette).

### Professional email structure

1. **Subject:** Clear — `CCC Assignment Unit 7 – Ravi Kumar`
2. **Greeting:** `Respected Sir/Madam` or `Dear [Name]`
3. **Body:** Short paragraphs; purpose in first lines
4. **Closing:** `Thank you` , `Regards` , your full name
5. **Signature:** Class, roll number, phone (if official)

### Do's and Don'ts

| Do | Don't |
|---|---|
| Use proper grammar | Write ALL CAPS (jorse chillana — looks rude) |
| Reply within reasonable time | Send virus attachments |
| Proofread before Send | Share passwords or OTP in email |
| Use BCC for bulk mail | Forward chain mails without checking |
| Keep official tone | Use slang in job/college mail |

### Spam and phishing in inbox

- **Spam** = unwanted bulk mail — mark as Spam, do not reply
- **Phishing email** = fake bank/Gmail warning — check sender address, do not click suspicious links

---

## 5. Online Meetings — Google Meet and Zoom

Online classes and interviews use **video conferencing** tools.

### Google Meet

- Join via **link** or **meeting code** from teacher
- Needs Google account or guest name entry
- Features: camera, mic, chat, screen share, raise hand
- **Mute** when not speaking (background noise kam)

### Zoom

- Join via link or Meeting ID + Passcode
- Host can control mute, waiting room, recording
- Chat panel for questions

### Meeting etiquette (CCC lab useful)

| Rule | Reason |
|---|---|
| Join 2–5 minutes early | Test mic and camera |

| | |
|---|---|
| Dress decently | Professional appearance |
| Stable internet | Avoid disconnect |
| Quiet place | Clear audio |
| Camera on when asked | Better engagement |
| Use chat for doubts | Do not interrupt speaker |

### Screen sharing

Teacher shares PPT; student may share assignment for demo. Only share intended window — close personal tabs first.

---

## 6. Cloud Storage and Collaboration

**Cloud** = files stored on internet servers, accessible from any device.

### Google Drive

- Free storage (15 GB with Google account)
- **Upload** files/folders from PC
- **Share** with link — permission: Viewer / Commenter / Editor
- **Google Docs, Sheets, Slides** — real-time collaboration (sah-karya)

### Microsoft OneDrive

- Linked to Microsoft account
- Integrates with Word, Excel online
- Share and co-edit documents

### Collaboration features

- **Version history** — see who changed what
- **Comments** — feedback without editing main text
- **Shared folder** — group project files in one place

### CCC exam points

- Cloud = backup + sharing + teamwork
- Link sharing safer than huge email attachments
- Set permission to **View only** when sending notes publicly

---

## 7. Social Media Basics

**Social media** = platforms to connect, share, and learn.

| Platform | Main use |
|---|---|
| WhatsApp | Messaging, groups, voice/video call |
| Facebook | Community, pages, events |

| | |
|---|---|
| Instagram | Photos, reels, short video |
| LinkedIn | Professional network, jobs |
| YouTube | Learning videos, tutorials |
| X (Twitter) | News, short updates |

### Responsible social media use

- Do not share **Aadhaar, OTP, bank details** in chat
- Verify news before forwarding (fake news / galat khabar)
- Respect privacy — ask before posting others' photos
- Report abuse using platform **Report** button
- Follow age rules and parental guidance

### WhatsApp for study groups

- Mute noisy groups during study time
- Use **broadcast lists** for one-way announcements
- Pin important messages (exam date, link)

---

## 8. Other Collaboration Tools (Awareness)

| Tool | Use |
|---|---|
| Microsoft Teams | College/office classes and chat |
| Slack | Workplace team channels |
| Telegram | Large channels, file sharing |
| Google Classroom | Assignments and grading |

CCC may ask concept — "tool for online class collaboration" → Meet, Zoom, Teams.

---

## 9. Important CCC Exam Terms

| Term | Meaning |
|---|---|
| Email | Electronic mail |
| CC | Carbon Copy — visible to all |
| BCC | Blind Carbon Copy — hidden |
| Attachment | File with email |
| Webmail | Email through browser |
| Cloud storage | Online file storage |
| Netiquette | Internet manners |

| | |
|---|---|
| Phishing | Fake messages to steal data |
| Screen share | Show your screen in meeting |
| 2FA / 2-Step | Extra login security |

---

## Lab Tasks

1. **Gmail setup:** Create or open Gmail — complete profile and recovery phone.
2. **Compose:** Send email to trainer with subject `CCC Unit 7 – [Your Name]` and short introduction.
3. **CC/BCC:** Send one mail with CC to classmate; another with BCC to two emails — observe visibility difference.
4. **Attachment:** Attach a Word/PDF assignment; note size limit message if file is large.
5. **Reply chain:** Reply and Reply All to a sample thread — note who receives mail.
6. **Forward:** Forward an institute notice to your own alternate email.
7. **Drive:** Upload assignment to Google Drive; share **view-only** link in email instead of attachment.
8. **Meet/Zoom:** Join a practice meeting — test mic, camera, chat, and mute.
9. **Netiquette:** Write one formal and one informal email on same topic — compare tone.
10. **Social media:** List 3 privacy settings you should enable on any one platform you use.

---

## Summary Checklist

- [ ] I can compose, reply, forward, and attach files in Gmail
- [ ] I know difference between To, CC, and BCC
- [ ] I follow netiquette for study and official mail
- [ ] I can join Meet/Zoom and use mute, chat, screen share
- [ ] I can upload and share files on Google Drive with correct permission
- [ ] I understand safe and responsible social media use
- [ ] I can identify spam and phishing emails

---

## Quick-Check MCQs (Questions Only)

1. Correct email format is (a) user@domain.com (b) user.domain only (c) @domain without user (d) http://user.com

2. CC in email means (a) visible copy to additional recipients (b) hidden copy (c) delete copy (d) print command

3. BCC is used when (a) recipients should not see each other's addresses (b) mail must be printed (c) attachment is removed (d) subject is blank

4. Paperclip icon in email is for (a) attachment (b) delete mail (c) change font (d) shutdown PC

5. Gmail is an example of (a) webmail service (b) operating system (c) spreadsheet (d) printer driver

6. Google Meet is mainly used for (a) online video meetings (b) virus scanning (c) disk formatting (d) BIOS setup

7. Cloud storage like Google Drive stores files (a) on internet servers accessible online (b) only inside CPU (c) on printer drum (d) in keyboard
8. Netiquette refers to (a) proper behaviour on the internet (b) network cable colour (c) antivirus brand (d) type of RAM
9. Sharing a 50 MB video in Gmail is better done by (a) Drive link instead of direct attachment (b) sending 50 separate emails (c) posting password in subject (d) disabling internet
10. Phishing email often tries to (a) steal login or bank details through fake messages (b) improve computer speed (c) install legal updates only (d) teach typing skills

# Subject 8: Digital Financial Services (6 Hours)

## Learning Objectives

- Understand digital payment systems used in India (UPI, wallets, net banking)
- Explain NPCI, BHIM, and AEPS in simple terms
- Use QR codes and mobile banking safely
- Know IMPS, NEFT, and RTGS for money transfer
- Follow OTP, PIN, and password safety for financial apps
- Identify common digital payment frauds and protections

---

## 1. Digital India and Cashless Economy

**Digital financial services** = banking and payments through mobile, internet, and cards instead of only cash (nakad).

### Benefits

- Fast transfer anytime (24×7 for many services)
- Less need to carry cash
- Easy record of transactions (ledger / lekhha)
- Government schemes and subsidies direct to bank (DBT — Direct Benefit Transfer)

### RBI and NPCI role

- **RBI (Reserve Bank of India)** — central bank; regulates banking and payment rules
- **NPCI (National Payments Corporation of India)** — builds and runs retail payment systems (UPI, RuPay, AePS)

---

## 2. UPI — Unified Payments Interface

**UPI** = instant bank-to-bank payment using mobile app and **UPI ID** or **QR code**.

### How UPI works (simple steps)

1. Install UPI app (PhonePe, Google Pay, Paytm, BHIM, bank app)
2. Link **bank account** and verify mobile number registered with bank
3. Create **UPI PIN** (4 or 6 digit — different from app login password)
4. Pay using UPI ID, QR scan, or mobile number

### UPI terms

| Term | Meaning |
|---|---|
| UPI ID / VPA | Virtual address like `name@oksbi` |
| UPI PIN | Secret PIN to authorize payment from bank |
| QR Code | Scannable code for merchant payment |

| | |
|---|---|
| Request money | Ask someone to pay you via UPI |
| Transaction ID | Reference number after payment — save screenshot |

### UPI limits (general awareness)

- Per transaction and per day limits set by bank and app
- For high amounts, bank may ask extra verification

---

## 3. BHIM App

**BHIM (Bharat Interface for Money)** = government-promoted UPI app by NPCI.

### Features

- Send/request money via UPI
- Scan QR and pay
- Check balance (with UPI PIN)
- Works across participating banks
- Available in multiple Indian languages

**Exam tip:** BHIM is based on **UPI platform** developed by NPCI.

---

## 4. Mobile Wallets

**Wallet** = stored prepaid balance in app for quick small payments.

| Wallet type | Example | Note |
|---|---|---|
| Semi-closed | Paytm wallet, PhonePe wallet | Pay approved merchants |
| Bank-linked UPI | Google Pay, BHIM | Direct from bank account |
| Open wallet | Some bank wallets | Broader use with KYC |

### Wallet vs UPI

- **Wallet:** Money loaded in advance in app
- **UPI:** Money debited directly from linked bank account

### KYC (Know Your Customer)

Identity verification (Aadhaar, PAN) for higher limits and full features — required by RBI rules.

---

## 5. Net Banking (Internet Banking)

**Net banking** = access bank account through bank website or app.

### Common services

- Check balance and mini statement
- Fund transfer (NEFT, IMPS, RTGS, within bank)

- Pay bills (electricity, mobile, tax)
- Open FD/RD, request cheque book
- Block lost debit card

### Login security

- **User ID** + **Password** + often **OTP** to registered mobile
- Use official bank URL — type yourself, do not click email links
- Enable transaction alerts on SMS/email

---

## 6. AePS — Aadhaar Enabled Payment System

**AePS** = withdraw cash, check balance, or deposit using **Aadhaar number** and **fingerprint** at bank mitra / micro ATM.

### Uses in rural India

- Banking without smartphone
- Government benefit withdrawal
- NPCI infrastructure through bank BC (Business Correspondent) points

**Exam point:** AePS uses Aadhaar authentication + bank link — operated under NPCI.

---

## 7. QR Code Payments

**QR (Quick Response) code** = square barcode scanned by phone camera.

### Types

- **Bharat QR** — standard for Indian merchants
- **UPI QR** — pay to merchant UPI ID
- **Static QR** — same code for all amounts (shop display)
- **Dynamic QR** — amount embedded for each bill

### Safe QR scanning

- Check merchant name shown in app before entering UPI PIN
- Do not scan unknown QR stickers in public places (fraud trap)
- Verify amount on screen matches bill

---

## 8. OTP Safety and Payment Security

**OTP (One Time Password)** = short code sent by bank/app for one transaction or login.

### Golden rules (CCC important)

| Rule | Why |
| --- | --- |
| Never share OTP with anyone | Bank never asks OTP on phone call |
| Do not share UPI PIN | PIN = full account access |

| | |
|---|---|
| Read SMS carefully | Check amount and payee name |
| Use official apps only | Fake apps steal credentials |
| Log out after use on shared phone | Protect account |
| Enable app lock / biometric | Extra layer |

### Common frauds (jaal / dhokha)

- **Fake customer care** number on Google — calls and asks OTP
- **Screen sharing** request during "refund"
- **Wrong QR** pasted over real merchant QR
- **Phishing links** — `bank-verify.fake.com`

**Report:** National Cyber Crime Portal — **cybercrime.gov.in**; bank helpline immediately.

---

## 9. Fund Transfer Methods — IMPS, NEFT, RTGS

| System | Speed | Hours | Typical use |
|---|---|---|---|
| **IMPS** | Instant (minutes) | 24×7 | Urgent small/medium transfer |
| **NEFT** | Batch — may take hours | Scheduled slots (now mostly 24×7) | Regular transfers |
| **RTGS** | Real-time large transfer | Banking hours (high value) | Big business payments |

### Details students should know

- **IMPS** — Immediate Payment Service — mobile/internet/net banking
- **NEFT** — National Electronic Funds Transfer
- **RTGS** — Real Time Gross Settlement — usually minimum amount limit (high)
- All need correct **account number** and **IFSC code**

**IFSC** = Indian Financial System Code — identifies bank branch (11 characters).

---

## 10. Cards and Other Digital Tools (Overview)

| Tool | Use |
|---|---|
| Debit card | Pay from savings account; ATM cash |
| Credit card | Borrow limit from bank — repay bill |
| RuPay card | Indian card network by NPCI |
| USSD *99# | Basic banking on simple mobile without internet |
| BBPS | Bharat Bill Payment System — unified bill pay |

---

## Lab Tasks

1. **UPI app:** Install BHIM or Google Pay — link bank in training/demo mode; note steps.
2. **UPI ID:** Find your UPI ID in app settings; write format `name@bankhandle`.
3. **QR scan:** Scan a sample merchant QR (trainer/demo) — observe name and amount screen before PIN.
4. **Net banking tour:** Login to bank demo site — locate balance, mini statement, fund transfer menu.
5. **Transfer types:** List when you would choose IMPS vs NEFT from bank help page.
6. **IFSC:** Find IFSC of your local branch from bank website or cheque leaf.
7. **OTP rules:** Write 5 situations when you must NOT share OTP.
8. **Fraud case study:** Read one RBI/NPCI awareness poster on UPI fraud — summarize 3 tips.
9. **AePS:** Watch demo video of Aadhaar fingerprint banking — list required items.
10. **Receipt habit:** After any demo payment, note transaction ID and screenshot practice.

---

## Summary Checklist

- [ ] I can explain UPI, BHIM, NPCI, and AePS
- [ ] I know difference between wallet and UPI bank transfer
- [ ] I can describe safe QR and UPI PIN practices
- [ ] I never share OTP or PIN with anyone on call or chat
- [ ] I know IMPS, NEFT, RTGS and IFSC purpose
- [ ] I use only official bank/app links for net banking
- [ ] I know where to report digital payment fraud

---

## Quick-Check MCQs (Questions Only)

1. UPI full form is (a) Unified Payments Interface (b) Universal Payment Internet (c) United Pay India (d) User Personal ID

2. NPCI develops (a) Indian retail payment systems like UPI (b) computer monitors (c) MS Word (d) printer drivers

3. BHIM app is based on (a) UPI platform (b) only email (c) floppy disk format (d) BIOS settings

4. UPI PIN is used to (a) authorize payment from linked bank account (b) unlock computer case (c) format hard disk (d) change monitor resolution

5. OTP should be (a) kept secret and never shared (b) posted on social media (c) shared with caller claiming to be bank (d) written on ATM wall

6. AePS uses (a) Aadhaar authentication for banking services (b) only email password (c) floppy disk (d) scanner calibration

7. IMPS transfer is generally (a) immediate and available 24×7 (b) only by post office (c) without any bank (d) only for foreign currency

8. IFSC code identifies (a) bank branch in electronic transfer (b) internet speed (c) printer model (d) keyboard layout

9. Scanning unknown QR code in public may lead to (a) payment fraud or wrong payee (b) free government laptop guarantee (c) automatic antivirus upgrade (d) faster CPU clock

10. Net banking should be accessed through (a) official bank website or app typed manually (b) random link in unknown SMS (c) public shared password (d) cyber café without logout

# Subject 9: e-Governance Services (6 Hours)

## Learning Objectives

- Explain e-governance and Digital India mission
- Use DigiLocker and UMANG for government services
- Understand online application process for PAN, passport, voter ID
- Verify official government websites (.gov.in, .nic.in)
- Fill online forms safely and save acknowledgements
- Know benefits of digital services for citizens (nagrik seva)

---

## 1. What is e-Governance?

**e-Governance (electronic governance)** = government services delivered through internet and digital platforms — fast, transparent, less paperwork (kam kagaz-kachhra).

### Traditional vs e-Governance

| Traditional | e-Governance |
|---|---|
| Long queues at office | Apply from home/mobile |
| Paper files | Digital records |
| Repeated visits | Track status online |
| Middlemen risk | Direct citizen–government link |

### Benefits for citizens

- **Transparency** — status tracking, less corruption opportunity
- **Speed** — faster processing when system works well
- **Accessibility** — rural areas via CSC, mobile apps
- **Cost saving** — travel and time (samay aur paise bachat)

---

## 2. Digital India Mission

**Digital India** = flagship programme to transform India into digitally empowered society.

### Nine pillars (awareness level)

1. Broadband Highways
2. Universal Access to Mobile Connectivity
3. Public Internet Access Programme
4. e-Governance — Reforming Government through Technology
5. e-Kranti — Electronic delivery of services
6. Information for All
7. Electronics Manufacturing
8. IT for Jobs

9. Early Harvest Programmes

### Key initiatives students should know

| Initiative | Purpose |
| --- | --- |
| Digital India | Overall digital empowerment |
| DigiLocker | Store official documents digitally |
| UMANG | One app for many government services |
| MyGov | Citizen participation platform |
| CSC (Common Service Centre) | Village-level digital seva kendra |
| UPI / DBT | Direct benefit to bank accounts |
| Aadhaar | Unique identity for many services |

---

## 3. DigiLocker

**DigiLocker** = secure cloud locker from Government of India to store and share verified documents.

**Website:** `https://www.digilocker.gov.in`

### Features

- Link with **Aadhaar** and mobile OTP
- Pull certificates from issuing boards (CBSE, etc.) if integrated
- Upload PDF documents (size limits apply)
- **eSign** documents digitally
- Share document link to employer/college with consent

### Documents commonly stored

- Marksheets, certificates
- Driving licence (if issued digitally)
- Vehicle registration (where available)
- PAN verification record

### Benefits

- No need to carry original everywhere
- Reduces fake document use when verified from issuer
- Available 24×7 on mobile app and web

---

## 4. UMANG App

**UMANG (Unified Mobile Application for New-age Governance)** = single mobile app for multiple government department services.

### Features

- One login for many services
- Available in Indian languages
- Bill pay, EPFO, PAN, passport enquiry, CBSE results (service list changes — check app)
- Push notifications for application status

**Exam tip:** UMANG is **integrated** platform — not replacing each department database but providing unified access.

---

## 5. Verifying Official Government Websites

Fake websites steal data and fees. Always **verify before pay or login**.

### Trusted domain patterns (India)

| Domain | Usually indicates |
|---|---|
| `.gov.in` | Government of India / state government |
| `.nic.in` | NIC-hosted government sites |
| Official department URL | Listed on India.gov.in or ministry site |

### Verification checklist

1. **HTTPS** padlock in browser
2. Domain spelling — `gov.in` not `gov.com` or `g0v.in`
3. Cross-check link from **official pamphlet** or India.gov.in
4. Never pay on site reached only from random WhatsApp forward
5. Look for **copyright / ministry name** and contact details

### Examples of official portals (awareness)

- Income Tax — incometax.gov.in
- Passport — portal2.passportindia.gov.in
- Voter — voters.eci.gov.in
- MCA / company — mca.gov.in
- DigiLocker — digilocker.gov.in

---

## 6. PAN Card — Online Services

**PAN (Permanent Account Number)** = tax identity issued by Income Tax Department.

### Online services (through NSDL or UTIITSL portals linked from incometax.gov.in)

- Apply for new PAN
- Correction / reprint
- Track application status
- Link Aadhaar with PAN (mandatory for many cases)

### Safe application steps

1. Open site only from **incometax.gov.in** links

2. Keep Aadhaar, photo, signature scan ready
3. Pay fee only on official payment gateway
4. Save **acknowledgement number** and receipt PDF
5. Track status using acknowledgement

---

## 7. Passport — Online Application

**Passport Seva** — online appointment and form system under Ministry of External Affairs.

### Process overview

1. Register on official Passport Seva portal
2. Fill online application form — choose fresh / reissue
3. Pay fee online
4. Book **Police Station / PSK appointment**
5. Visit centre with original documents
6. Track application status online

### Documents (typical — verify latest rules)

- Aadhaar, birth proof, address proof, photos
- Old passport if reissue

**Important:** Only use **passportindia.gov.in** official links — agents may charge extra unnecessarily.

---

## 8. Voter ID — Online Services

**Election Commission of India (ECI)** provides online voter services.

**Portal:** voters.eci.gov.in (verify current URL on ECI site)

### Services include

- New voter registration (Form 6)
- Correction of details (Form 8)
- Migration / address change
- Download e-EPIC (digital voter card where available)
- Search name in electoral roll

### Steps awareness

- Fill Form online with mobile OTP
- Upload supporting documents if required
- Note **reference number**
- Track application status

Voting is both **right and duty** (matadhikar aur kartavya) — keep voter details updated.

---

## 9. Other Common e-Governance Services

| Service | Portal / App | Use |
|---|---|---|

| | | |
|---|---|---|
| Aadhaar update | uidai.gov.in | Address/mobile update, download |
| Driving licence | Parivahan / state transport | Apply, renewal |
| Birth/death certificate | State municipal portals | Civil registration |
| PM schemes | respective ministry sites | Scholarship, pension enquiry |
| GST (business) | gst.gov.in | Business taxpayers |
| FASTag / NHAI | nhai.gov.in | Toll payments |

### CSC — Common Service Centre

Village entrepreneur runs CSC with computer — helps citizens who cannot use internet alone. Part of Digital India last-mile delivery.

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## 10. Online Form Best Practices

### Before filling

- Read **instructions PDF** completely
- Keep scanned documents correct size (KB/MB limit)
- Stable internet and power backup

### While filling

- Match spelling with **Aadhaar and certificates**
- Save draft if option available
- Use valid email and mobile — OTP will come there

### After submit

- Save **acknowledgement / application ID** screenshot
- Print or PDF confirmation page
- Note helpline number from official site only

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## Lab Tasks

1. **Digital India:** Visit digitalindia.gov.in — list 3 objectives of the mission.
2. **DigiLocker:** Create account (or demo walkthrough) — explore issued documents and upload sections.
3. **UMANG:** Install UMANG — browse 5 available services in your state.
4. **Domain check:** Compare real `.gov.in` site vs a fake-looking URL — write 4 differences.
5. **PAN awareness:** Open incometax.gov.in — locate PAN services link (do not submit fake data).
6. **Passport:** Explore Passport Seva site — list documents needed for fresh passport (from official FAQ).
7. **Voter:** Open ECI voter portal — find "search in electoral roll" or registration form name.
8. **Form practice:** Fill a sample HTML practice form with validation (trainer provided).
9. **Acknowledgement:** Document what to save after submitting any government online form.
10. **CSC:** Find nearest CSC on locator (if available) — note services offered.

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## Summary Checklist

- ☐ I can explain e-governance and Digital India benefits
- ☐ I know DigiLocker and UMANG main purpose
- ☐ I can verify official .gov.in / .nic.in websites
- ☐ I understand online PAN, passport, voter application flow
- ☐ I save acknowledgement and track application status
- ☐ I avoid fake government websites and phishing
- ☐ I know CSC helps rural digital access

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## Quick-Check MCQs (Questions Only)

1. e-Governance means (a) delivering government services through digital means (b) removing all government offices (c) only playing online games (d) banning internet in villages

2. DigiLocker is used to (a) store and share verified digital documents (b) download movies only (c) format computer (d) repair printer

3. UMANG app provides (a) unified access to multiple government services (b) only social media chat (c) hardware drivers (d) BIOS update

4. Official Indian government sites often use domain (a) .gov.in (b) .com.only.fake (c) .freegames (d) .personal.blog

5. Digital India mission aims to (a) digitally empower citizens and improve governance (b) stop all cash use overnight without technology (c) remove mobile phones (d) close all schools

6. PAN card is related to (a) Income Tax identity (b) driving licence only (c) voter list only (d) passport immigration only

7. Passport Seva online portal helps in (a) application and appointment booking (b) CPU upgrade (c) monitor repair (d) keyboard cleaning

8. Voter registration online in India is managed under (a) Election Commission of India (b) private shopping mall (c) game company (d) local cable operator only

9. Before paying on a government website you should (a) verify official URL and HTTPS (b) trust any WhatsApp link (c) share OTP with helper (d) ignore spelling of domain

10. CSC centres in villages mainly provide (a) assisted digital government and business services (b) only coffee shop (c) computer hardware manufacturing (d) antivirus CD sales only

# Subject 10: Cyber Security (6 Hours)

## Learning Objectives

- Define virus, malware, phishing, ransomware, and spyware
- Create and manage strong passwords
- Understand Two-Factor Authentication (2FA)
- Explain firewall, antivirus, and safe browsing
- Recognize social engineering and online fraud
- Know how to report cybercrime at cybercrime.gov.in
- Build daily habits for personal and device security

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## 1. What is Cyber Security?

**Cyber security** = protecting computers, networks, data, and accounts from digital attacks (cyber hamle se suraksha).

### Why it matters for CCC students

- Daily use of Gmail, UPI, social media
- Exams and jobs online
- Personal photos and documents on phone
- One mistake can cause money loss or identity theft (pehchaan chori)

### CIA triad (basic concept)

| Principle | Meaning |
|---|---|
| **Confidentiality** | Only authorized person sees data |
| **Integrity** | Data is not changed wrongly |
| **Availability** | System works when needed |

---

## 2. Malware — Malicious Software

**Malware** = any software designed to harm or steal.

### Common types

| Type | What it does | How it spreads |
|---|---|---|
| **Virus** | Attaches to file/program; spreads when opened | USB, email attachment, infected download |
| **Worm** | Spreads alone over network | Weak passwords, unpatched systems |
| **Trojan** | Looks like useful software; hidden harm | Fake game, cracked app |
| **Spyware** | Secretly watches activity, keystrokes | Bundled with free software |

| | | |
|---|---|---|
| **Ransomware** | Locks files; demands money to unlock | Phishing links, bad attachments |
| **Adware** | Unwanted ads; may track browsing | Free toolbars, shady installers |

### Signs of infection

- PC very slow suddenly
- Unknown programs start automatically
- Files missing or extension changed ( `.locked` )
- Antivirus disabled without your action
- Unexpected pop-ups and browser homepage change

---

## 3. Phishing and Social Engineering

**Phishing** = fake message or website pretending to be bank, Gmail, or government to steal password/OTP.

### Phishing channels

- Email — "Your account will close, click here"
- SMS / WhatsApp — fake KYC, lottery win, parcel link
- Phone call — "I am from bank, tell OTP"
- Fake websites — copy of SBI/Gmail login page

### Social engineering

Tricking human mind (manav ko dhokha) instead of breaking code:

- Urgency — "Act in 10 minutes"
- Fear — "IT raid, pay now"
- Greed — "You won ₹50 lakh"
- Authority — "I am police / RBI officer"

### How to avoid

- Never share **OTP, UPI PIN, password, CVV**
- Type bank URL yourself
- Check sender email address carefully
- Call official helpline from bank card — not from SMS number

---

## 4. Ransomware Awareness

**Ransomware** encrypts (locks) your files and criminals ask **ransom** (firauti) in crypto or UPI.

### Protection

- **Backup** important files to external drive or cloud regularly
- Do not open unknown attachments — especially `.zip` , `.exe`
- Keep Windows and apps **updated**
- Use standard user account — not always Administrator

### If attacked

- Do not pay immediately — payment does not guarantee unlock

- Disconnect from network
- Report to police / cybercrime.gov.in
- Restore from backup if available

---

## 5. Strong Passwords

**Password** = first lock on your digital life.

### Weak passwords (never use)

- `123456`, `password`, your name, birth date, `qwerty`
- Same password on Gmail and bank

### Strong password rules

- **8+ characters** (12+ better)
- Mix **uppercase, lowercase, numbers, symbols**
- Unique for each important account
- Not guessable from Facebook info

### Example technique — passphrase

`Mera@CCC2025$Safe!` — long phrase with symbols (do not use this exact one publicly)

### Password managers (awareness)

Apps like built-in browser manager or Bitwarden store complex passwords securely.

### If password leaked

Change immediately; enable 2FA; check bank transactions.

---

## 6. Two-Factor Authentication (2FA)

**2FA / Two-Step Verification** = two proofs needed to login:

1. Something you **know** — password
2. Something you **have** — OTP on phone, authenticator app, or security key

### Where to enable (must for students)

- Gmail / Google account
- Banking and UPI apps
- Social media (Instagram, Facebook)
- DigiLocker, income tax portal

### OTP types

- SMS OTP (common; SIM swap risk — keep SIM safe)
- Email OTP
- Authenticator app (Google Authenticator) — works without SMS

**Exam point:** 2FA improves security even if password is stolen.

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## 7. Firewall

**Firewall** = barrier controlling incoming and outgoing network traffic.

### Types

- **Windows Firewall** — built into Windows; block unauthorized access
- **Hardware firewall** — in router
- **Corporate firewall** — college/office network

### What it does

- Blocks suspicious connection requests
- Allows safe browsing when configured properly
- Not a substitute for antivirus — use **both**

---

## 8. Antivirus and Anti-malware

**Antivirus** = software detecting and removing malware.

### Popular examples

Windows Security (Defender), Avast, Kaspersky, Bitdefender, Norton

### Good practices

- Keep virus definitions **updated**
- Run **full scan** weekly on home PC
- Scan USB before opening files
- Download only from official sites

### Limitations

- Cannot stop you from giving OTP to scammer
- May not catch newest threats — behaviour matters

---

## 9. Safe Browsing Habits

| Habit | Benefit |
|---|---|
| Use HTTPS on login/payment | Encrypted connection |
| Log out on shared PC | Others cannot open your account |
| Update browser and OS | Security patches |
| Avoid public Wi-Fi for banking | Reduce interception risk |
| Check app permissions on phone | Stop unnecessary access to contacts/camera |
| Review privacy settings on social media | Limit public data |
| Think before click | Stop phishing |

### Physical security

- Lock phone with PIN/fingerprint
- Do not leave laptop unlocked in lab
- Shred or secure papers with account numbers

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## 10. Reporting Cybercrime in India

**National Cyber Crime Reporting Portal:** `https://www.cybercrime.gov.in`

### What you can report

- Online financial fraud
- Social media harassment
- Hacking, data theft
- Child-related offences online (priority)
- Ransomware, sextortion

### How to report (overview)

1. Visit cybercrime.gov.in
2. File complaint — **Cyber Crime Reporting** section
3. Provide evidence — screenshots, URLs, numbers, transaction IDs
4. Note **complaint reference number**
5. Also inform bank if money lost — block transaction quickly

### Helpline

**1930** — Cyber fraud financial helpline (many states — verify local availability)

### Evidence to preserve

- SMS, email headers
- UPI transaction screenshot
- Caller number and time
- WhatsApp chat export if needed

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## 11. CCC Exam Quick Threat Summary

| Threat | One-line definition |
|---|---|
| Virus | Malware that spreads via files |
| Phishing | Fake messages to steal secrets |
| Spam | Unwanted bulk messages |
| Spyware | Hidden monitoring software |
| Ransomware | Locks data for money |
| Trojan | Disguised harmful program |

| | |
|---|---|
| Firewall | Traffic filter for network |
| 2FA | Two-step login security |
| Identity theft | Using your identity fraudulently |

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## Lab Tasks

1. **Password audit:** Check if you reuse passwords — create 3 strong unique passphrases on paper (private, not shared).
2. **2FA:** Enable 2-Step Verification on Google account — note backup codes storage.
3. **Windows Security:** Open Windows Security — run quick scan; check firewall is ON.
4. **Phishing examples:** Trainer shows 3 sample fake SMS/emails — identify red flags.
5. **URL check:** Compare real vs fake login URL — list 5 differences.
6. **Browser:** Review privacy settings — cookies, safe browsing, password save option.
7. **USB caution:** Discuss why autorun USB from unknown person is risky.
8. **Backup:** Copy important folder to cloud or pen drive — document backup date.
9. **Report portal:** Open cybercrime.gov.in — explore complaint categories (do not file test fake complaint).
10. **Security pledge:** Write personal 10-point cyber safety rules for home and lab.

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## Summary Checklist

- [ ] I can define virus, malware, phishing, ransomware, spyware
- [ ] I create strong unique passwords and enable 2FA
- [ ] I understand firewall and antivirus roles
- [ ] I follow safe browsing and phone lock habits
- [ ] I never share OTP, PIN, or password with anyone
- [ ] I know how to report at cybercrime.gov.in and helpline 1930
- [ ] I keep backups and Windows/apps updated

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## Quick-Check MCQs (Questions Only)

1. Virus is a type of (a) harmful software that can spread (b) hardware printer part (c) network cable (d) keyboard cleaning liquid

2. Phishing attack mainly tries to (a) trick user into revealing sensitive information (b) increase internet speed (c) format monitor (d) improve typing accuracy

3. Ransomware typically (a) locks files and demands payment (b) gives free antivirus (c) increases RAM automatically (d) teaches coding

4. Strong password should (a) mix letters, numbers, symbols and be 8+ characters (b) be your name only (c) be 123456 for easy recall (d) same for all websites

5. 2FA means (a) two different steps to verify login (b) two printers connected (c) two monitors only (d) second email account for spam

6. Firewall is used to (a) monitor and control network traffic (b) clean monitor screen (c) charge mobile battery (d) print documents faster

7. Antivirus software helps to (a) detect and remove malware (b) create viruses (c) slow CPU intentionally (d) replace internet connection
8. OTP shared with unknown caller may lead to (a) financial fraud and account takeover (b) free government laptop (c) automatic scholarship (d) faster broadband
9. Cybercrime in India can be reported at (a) cybercrime.gov.in portal (b) random WhatsApp group (c) unknown .com lottery site (d) pirated software forum
10. Regular data backup protects against (a) ransomware and accidental file loss (b) slow mouse only (c) keyboard layout change (d) monitor brightness issues

# Subject 11: Future Skills & Digital Literacy (4 Hours)

## Learning Objectives

- Understand basics of Artificial Intelligence (AI) and Machine Learning (ML)
- Explain IoT, cloud computing, and their daily-life examples
- Define digital literacy and its importance for jobs
- Get introductory awareness of coding and computational thinking
- Discuss technology ethics, privacy, and responsible AI use
- Develop mindset for lifelong learning (jeevan-anty siksha)

---

## 1. Digital Literacy in the 21st Century

**Digital literacy** = ability to use digital tools safely and effectively to learn, work, and participate in society (digital duniya mein safal rehna).

### Components

| Skill area | Examples |
|---|---|
| Basic operations | Smartphone, browser, email, UPI |
| Information literacy | Search, verify fake news |
| Communication | Email etiquette, online meetings |
| Safety | Passwords, privacy, cyber awareness |
| Problem solving | Try settings, help menu, restart |
| Creation | Documents, presentations, simple posts |

### Why CCC connects to future skills

CCC builds **foundation** — without basics, advanced tools are harder. Certificate shows you can function in digital workplace.

---

## 2. Artificial Intelligence (AI) — Basics

**AI (Artificial Intelligence)** = computer systems that perform tasks needing human-like intelligence — understanding language, recognizing images, making suggestions.

### AI in daily life (India examples)

- Google Maps traffic route suggestion
- YouTube recommended videos
- Phone face unlock
- Chatbots on bank and shopping sites
- Voice assistants — Google Assistant, Alexa

- UPI fraud detection patterns

### AI vs normal program

| Normal program | AI system |
| --- | --- |
| Fixed step-by-step rules | Learns patterns from data |
| Same output for same input usually | May improve with more data |
| Calculator, MS Word spell simple rule | Image recognition, language translation |

### Types of AI (awareness)

- **Narrow AI** — one task (chess, spam filter) — all current practical AI
- **General AI** — human-level all tasks — research stage, not common yet

**Hindi note:** AI = **kritrim buddhi** (machine showing smart behaviour).

---

## 3. Machine Learning (ML) — Introduction

**Machine Learning** = branch of AI where computer **learns from data** without being explicitly programmed for every rule.

### Simple example

Email spam filter:

- Trained on thousands of spam and good mails
- Learns words and patterns
- Classifies new mail as spam or not

### Common ML uses

- Recommendation (Netflix, Flipkart)
- Medical image assist (doctor support, not replacement)
- Crop disease detection apps for farmers
- Speech-to-text in Indian languages

### Data is fuel

More quality data → often better ML model.
**Privacy concern:** data must be collected with consent and protection.

---

## 4. Internet of Things (IoT)

**IoT** = everyday physical devices connected to internet, sending and receiving data.

### Examples

| Device | Smart behaviour |
| --- | --- |
| Smart watch | Heart rate, steps to phone |

| | |
|---|---|
| Smart bulb | Control brightness from app |
| CCTV IP camera | View home on mobile |
| Smart AC | Schedule temperature |
| GPS tracker on vehicle | Live location |
| Soil moisture sensor | Farm irrigation alert |

### IoT structure (simple)

**Sensor → Internet → App/Cloud → User action**

### Risks

- Weak default passwords on cameras
- Data leakage if company servers hacked
- Always change default password; update firmware

---

## 5. Cloud Computing — Recap for Future Context

**Cloud computing** = using computing services (storage, software, servers) over internet on rent — pay as you use.

### Service models (basic)

| Model | Meaning | Example |
|---|---|---|
| SaaS | Software as Service | Gmail, Google Docs |
| PaaS | Platform for developers | Heroku, Google App Engine |
| IaaS | Infrastructure (servers) | AWS EC2, Azure VMs |

### Benefits

- No heavy server in office
- Scale up during festival sale for business
- Access from anywhere

CCC already uses cloud — Drive, webmail. Future jobs may use cloud dashboards and online CRM tools.

---

## 6. Introduction to Coding

**Coding (programming)** = writing instructions in a language computers understand.

### Why learn basics even if not developer

- Automate boring tasks (Excel macros, simple scripts)
- Understand app limitations and bugs
- Better career in IT, data, testing, design, government tech posts
- Logical thinking (logical soch)

### Popular languages (awareness)

| Language | Common use |
|---|---|
| Python | AI, data science, beginners |
| JavaScript | Websites, apps |
| Java | Android, enterprise |
| C / C++ | System, competitive base |
| HTML/CSS | Web page structure and style |

### Computational thinking

1. **Decomposition** — break big problem into small parts
2. **Pattern recognition** — find similar problems
3. **Abstraction** — hide unnecessary detail
4. **Algorithm** — step-by-step solution

### Free learning paths

- Code.org, Khan Academy, SWAYAM MOOC courses
- YouTube Hindi/English beginner series
- NIELIT / CSC skill courses

**First program tradition:** Print `Hello World` — just displays greeting on screen.

---

## 7. Technology Ethics and Responsible Use

**Digital ethics** = right and wrong use of technology respecting people and law.

### Key topics

| Topic | Issue | Responsible approach |
|---|---|---|
| Privacy | Apps collect location, contacts | Read permissions; minimal sharing |
| Fake news | False information spreads fast | Verify before forward |
| AI bias | System unfair to some groups | Demand fair testing; diverse data |
| Deepfake | Fake video of real person | Do not believe everything; check source |
| Copyright | Copying videos, books illegally | Use licensed or free content |
| E-waste | Throwing old phones in trash | Recycle at authorized centre |
| Screen addiction | Health and study harm | Time limits, breaks |

### AI in exams and jobs

- Using ChatGPT to **learn** concepts — good if you verify
- Copy-paste exam answers — **cheating**; harms your skill

- Disclose AI use where institution requires

**Indian framework awareness:** NITI Aayog National Strategy for AI, Digital Personal Data Protection Act (DPDP) — privacy law context.

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## 8. Lifelong Learning (Continuous Upskilling)

Technology changes fast — today's tool may update next year.

### Habits for lifelong learning

- Learn **one new tool** each month (shortcut, app feature)
- Follow official blogs — Google, Microsoft Learn
- Practice **English + Hindi** technical terms for exams and jobs
- Join free webinars — Skill India, NSDC, NIELIT
- Build portfolio — certificates, small projects, volunteer digital seva

### Future job areas (digital)

- Data entry and office automation
- Digital marketing assistant
- Cyber security support
- IoT technician
- AI data labeling and testing
- Government e-governance operator at CSC

### Growth mindset (vriddhi soch)

Mistakes in lab are learning — ask teacher, search documentation, try again.

---

## 9. Emerging Trends (Short Awareness)

| Trend | One-line |
|---|---|
| 5G | Faster mobile internet for video and IoT |
| Blockchain | Distributed record — used in some certificates and finance |
| AR/VR | Augmented/Virtual reality for training and games |
| Robotics | Machines doing physical tasks in factory and home |
| Green IT | Energy-efficient data centres and e-waste control |
| BharatGPT / Indian language AI | Local language technology growth |

CCC exam may ask definition-level questions — not deep programming.

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## Lab Tasks

1. **AI hunt:** List 5 AI features you already use on phone — explain each in one line.
2. **ML example:** Discuss how YouTube or Flipkart recommendations might use past behaviour data.

3. **IoT home:** Name 3 IoT devices you have seen in shop or home — what data they may send.
4. **Cloud check:** Identify 3 SaaS apps you use daily (Gmail, WhatsApp Web, etc.).
5. **Code hello:** Open code.org or Python online editor — run Hello World (trainer demo).
6. **Logic puzzle:** Write step-by-step algorithm to make tea or login to Gmail — test with classmate.
7. **Fake news:** Take one viral message — verify on fact-check site or official source.
8. **Permissions:** Review mobile app permissions — disable one unnecessary access.
9. **Learn plan:** Write 6-month skill plan — CCC, typing, Excel, one MOOC course.
10. **Ethics debate:** Group discussion — "Should students use AI chatbots for homework?" — list pros and cons.

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## Summary Checklist

- [ ] I can explain AI, ML, IoT, and cloud in simple words with examples
- [ ] I understand digital literacy as broader than just using phone
- [ ] I know basic coding purpose and computational thinking steps
- [ ] I follow ethical and responsible technology use
- [ ] I verify information before sharing online
- [ ] I have a plan for continuous learning after CCC
- [ ] I connect future skills to jobs and Digital India goals

---

## Quick-Check MCQs (Questions Only)

1. Artificial Intelligence (AI) refers to (a) machines performing tasks needing human-like intelligence (b) only printing documents (c) cleaning hardware with water (d) typing speed only

2. Machine Learning is (a) learning patterns from data by computer (b) reading machine manual only (c) repairing printer drum (d) BIOS password reset

3. IoT means (a) everyday devices connected to internet sharing data (b) internet without any device (c) only email system (d) offline calculator

4. Digital literacy includes (a) safe and effective use of digital technology (b) ignoring cyber safety (c) only playing games (d) avoiding all learning apps

5. Cloud computing provides (a) online services like storage and software on demand (b) only rain water harvesting (c) physical-only floppy disks (d) CRT monitor repair

6. Python is often used for (a) beginners, AI and data tasks (b) only washing machine repair (c) BIOS chip replacement (d) mouse ball cleaning

7. Computational thinking includes (a) breaking problem into smaller steps (b) shouting at monitor (c) deleting all files (d) disabling all security

8. Sharing unverified viral news may cause (a) spread of misinformation (b) automatic job promotion (c) free laptop from government always (d) faster CPU clock speed

9. Lifelong learning means (a) continuously updating skills through life (b) stopping study after one exam (c) avoiding all new technology (d) only reading paper books forever

10. Using AI chatbot to copy entire exam answer without understanding is (a) unethical and harmful to real learning (b) always officially encouraged in all exams (c) required by CCC syllabus practical (d) same as

practical lab work